METRIC

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PERFORMANCE SPECIFICATION

BATTERIES, LITHIUM, RECHARGEABLE, AIRCRAFT, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for secondary (rechargeable) storage batteries of lithium electrochemistry including, but not limited to, lithium-ion, gel-polymer lithium-ion, and lithium polymer.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, System Standardization and PHS&T Branch, Code BL32600, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to LKE-NAVAIRDocComments@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-18148/3	-	Receptacles, Electric, Aircraft Storage Battery
MIL-PRF-24236	-	Switches, Thermostatic, (Metallic and Bimetallic), General Specification for
MIL-PRF-29595/1	-	Battery, Storage, Aircraft, Maintenance-Free, 26-Volt, 25-Ampere-Hour
MIL-PRF-29595/2	-	Battery, Storage, Aircraft, Maintenance-Free, 26-Volt, 50-Ampere-Hour, Lithium Rechargeable
MIL-DTL-38999	-	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect (Bayonet, Threaded, or Breech Coupling), Environment Resistant with Crimp Removable Contacts or Hermetically Sealed with Fixed, Solderable Contacts, General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202-302	-	Method 302, Insulation Resistance
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-464	-	Electromagnetic Environmental Effects Requirements for Systems
MIL-STD-704	-	Aircraft Electric Power Characteristics
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-704-1	-	Guidance for Test Procedures for Demonstration of Utilization Equipment Compliance to Aircraft Electrical Power Characteristics (Part 1 of 8 Parts)
MIL-HDBK-704-8	-	Guidance for Test Procedures for Demonstration of Utilization Equipment Compliance to Aircraft Electrical Power Characteristics 28 VDC (Part 8 or 8 Parts)

(Copies of these documents are available online at https://quicksearch.dla.mil.)

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

S9310-AQ-SAF-010 - Technical Manual for Batteries, Navy Lithium Safety Program Responsibilities and Procedures

(Copies of Navy Technical Manual S9310-AQ-SAF-010 are available from https://www.public.navy.mil/navsafecen/Documents/afloat/Surface/CS/Lithium_Batteries_Info/LithBattSafe.pdf/.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of this document are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME-Y14.38 - Abbreviations and Acronyms for Use in Product Definition and Related Documents

(Copies of this document are available from http://www.asme.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) INTERNATIONAL

SAE AMS-STD-595/17875 - Miscellaneous, Gloss

(Copies of this document are available from http://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Cells and batteries furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified products list (QPL) before contract award (see 4.2 and 6.4).

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3.3 Components. Components that are specified shall be in accordance with the applicable specification sheet. Components shall be chosen so that the battery meets the performance requirements of this specification and if requested by the qualifying activity, the manufacturer shall supply a certification of conformity of the components (see 6.2.e and 6.10).

3.4 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not contact each other (see 6.10.4).

3.5 Design. The design shall conform to the requirements specified herein when examined in accordance with the visual and mechanical tests of 4.5.2. The design shall also conform to the requirements specified in S9310-SAF-AQ-010 (see 6.18 and 6.19). Requirements for the individual types of batteries are specified in the applicable specification sheets (see 3.1).

3.5.1 General. The battery shall be free from defects that will affect life, functioning, and appearance. Batteries shall not have loose contacts, poor or improper molding or fabrication, damaged or improperly assembled contacts, peeling, flaking or chipping of plating or finish, mechanical damage due to testing environments, nicks or burrs of metal parts of surfaces, nor improper or incorrect marking.

3.5.2 Batteries. After delivery from the manufacturer, batteries shall not require the addition of any substance before being put into service. The battery shall not require any maintenance besides a conditioning charge, in meeting the provisions of this specification (see 6.15.8).

3.5.3 Battery container and cover. The battery container and cover shall be free of pits, blow holes, rough spots, and other deformations. The dimensions and locations of receptacles, hold-downs, vent tubes, and battery identification shall conform to the applicable specification sheet. The cover shall not be removable from the container. The battery container and cover shall neither support combustion nor emit toxic vapors when subjected to flame.

3.5.4 Venting. Each cell and battery shall incorporate a safety-venting device or be designed and manufactured in such a manner that will preclude a violent rupture as a result of cell venting. The design and construction shall not degrade or obstruct the vent. If a battery includes vent tubes, the tubes shall be located as shown on the applicable specification sheet. Each vent tube shall be capable of supporting the test sample such that the requirements of 3.9.5 are met after being tested in accordance with 4.5.6.

3.5.5 Receptacles. Each battery shall include a main power receptacle in accordance with the applicable specification sheet and shall be designated as J1. This receptacle provides the battery's interface to the aircraft or equipment electrical system. The hardness of the worm gear retaining pins within the sockets of receptacles in accordance with MIL-PRF-18148/3 receptacles shall be not less than 55 on the Rockwell C Scale when tested in accordance with 4.5.22. Each battery shall contain a second sensor receptacle and be designated as J2. The J2

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receptacle functions may include i) communication to the on-aircraft battery monitoring system, ii) provide power to and from the aircraft DC bus, external charger, and/or external off-aircraft maintenance equipment, iii) heater power input and return, and iv) data transmission in accordance with the applicable specification sheet. The J2 receptacle shall be in accordance with MIL-DTL-38999 and have a configuration as shown in the applicable specification sheet. Electroless nickel plating shall not be used on the shell of the J2 receptacle. The electrical schematic pin assignments or connector interface for the J2 receptacle shall be in accordance with the applicable specification sheet.

3.5.6 Connectors, intercell. All electrical connections within the battery shall be by surface-to-surface conduction and not through screw threads.

3.5.7 Leakage. Evidence of leakage of electrolyte or any other substance during the performance of any of the tests specified in section 4 shall not occur.

3.5.8 Voltage. Unless otherwise specified in the applicable specification sheet, the nominal potential of the battery shall be 26 volts DC. The battery potential shall be not greater than 29.0 volts at any time.

3.5.9 Heaters. Heaters may be incorporated into batteries to meet the cold temperature charging requirements. Heater power shall not be provided during any discharge performed in section 4. Heater power shall be provided during all constant potential charges (see 4.4.2). Heater power may be available prior to discharging the battery in accordance with the applicable specification sheet. De-rated power output from the battery at low temperatures may be allowable in accordance with the applicable specification sheet. When heaters are incorporated, the following requirements shall apply:

a. Heaters shall be powered by 115 Volts Alternating Current (VAC) or by 28.0 Volts Direct Current (VDC) (see applicable specification sheet for heater power requirements).

b. The rated current of each heater control device shall be compatible with the current specified in the applicable battery specification sheet.

c. Heater elements and wire assemblies shall be electrically insulated and installed in a manner which minimizes potential battery damage due to electrical shorting, sparking, or other electrical hazards.

d. All electrical switches, controls, circuits, and installations shall be explosion-proof. When operating, they shall not ignite explosive mixtures of hydrogen and oxygen gasses generated inside the battery container.

e. Both primary and secondary (backup) temperature limiting controls shall be incorporated into the battery's electrical circuits to prevent overheating and shall be wired in series. If the primary heater control fails, the battery shall not reach a damaging temperature.

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f. All thermal switches shall be snap-action type in accordance with MIL-PRF-24236.

g. All heater components shall be located inside the battery container.

h. When the battery is operating in accordance with any specified environmental condition (see 4.4.1), the exterior surface of the battery container shall not be greater than 75 °C (167 °F) due to the heater operation.

3.6 Physical requirements.

3.6.1 Atmospheric operating conditions. Unless otherwise specified in the applicable specification sheet, the cell or battery shall be capable of performing electrically while encountering the environments of 3.6.1.1 through 3.6.1.3.

3.6.1.1 Humidity. The operating relative humidity shall be from 0 percent to 100 percent including condensation when tested in accordance with 4.5.29.

3.6.1.2 Altitude. The operating atmospheric pressure shall range from that found at sea level to that found at an altitude of 18,288 meters (60,000 feet) when tested in accordance with 4.5.23 and 4.5.28.

3.6.1.3 Temperature. The operating atmospheric temperature shall be from -40° C to +71 °C (-40 °F to 160 °F).

3.6.2 Orientation. The battery shall be designed for operation in any position while meeting the capacity, voltage and electrolyte leakage requirements (see applicable specification sheet) when tested in accordance with 4.5.14.

3.7 Color and marking. The batteries shall conform to the requirements listed below when examined in accordance with 4.5.2.

3.7.1 Container color. The battery shall conform to one of the following color styles:

a. The color under natural light of four out of six faces of the battery, except the mounting points, hold-down hooks, other external hardware, identification marking, and instructions, shall conform to the color listed in Table I.

b. Color only the container cover and marking backgrounds in accordance with those listed in Table I under natural light; the rest of the container shall be white or grey.

3.7.2 Polarity marking. The container body shall be conspicuously and durably marked "+" in red in the location shown on the applicable specification sheet.

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3.7.3 Identification marking. The battery identification marking shall be in white or black print and shall be clear and legible throughout all tests. Unless otherwise specified in the applicable specification sheet, the marking shall use a type size of not less than 4.2 mm. The marking shall contain the required titles and information as shown in figure 1 herein and shall be placed in accordance with the applicable specification sheet. The lot code shall be constructed in accordance with 3.7.6. The blank area following the entry Date First Placed in Service shall be capable of being written on with a felt tip permanent marker. The manufacturer shall fill in the applicable information at the indicated areas. Abbreviations in accordance with ASME-Y14.38 and acronyms are permitted.

3.7.4 Battery caution marking. The battery caution marking shall contain the required titles and information as shown on figure 2 herein and shall be placed in accordance with the applicable specification sheet. The lettering height shall be not less than 6.3 mm. The marking shall be in black print.

3.7.5 Marking for batteries with heaters. If any exterior surface of the battery container exceeds 49 °C (120 °F) when all heater elements are energized and all other battery electrical circuits are not functioning for 1 hour, the following caution statement shall be inscribed on the battery identification marking and battery caution marking: CAUTION HOT SURFACE (see figures 1 and 2). The height of all statement letters shall be equal and not less than 6.2 mm. The battery container shall also display an electrical schematic for internal heaters. The height of the letters and numbers shall be equal and not less than 3.8 mm.

3.7.6 Lot code. The month and year that the manufacturer compiled the batteries into a group for inspection and the lot code of the battery shall be clearly shown on the battery identification marking. The information shall be coded such that the first two digits indicate the month of the year, the middle two digits indicate the year (followed by a dash), and the last two digits indicate the inspection lot (see 6.15.7) number. Months earlier than the tenth shall be indicated by a single digit preceded by "0". When the inspection lot is compiled during the first three or the last three working days of the month, the manufacturer is permitted to use either that month or the nearest adjacent month as the coded month. For example, lot code 0409-12 indicates the lot was compiled in April 2009 and was the twelfth lot.

3.7.7 Warnings. The cell or battery shall be marked with the following warnings and shall be placed in accordance with the applicable specification sheet. The warning shall be as follows:

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For Use in Designated Equipment Only

Caution

Lithium Ion Battery

Recharge in accordance with authorized/approved instructions

Do not pierce, cannibalize, mutilate, puncture, short circuit, expose to temperatures above 85 °C, or incinerate. Use in equipment approved for this cell or battery. Do not throw in trash.

3.8 System safety design requirements.

3.8.1 General safety requirements. Unless otherwise noted, the battery, when subjected to any tests specified in section 4, shall not show:

a. Explosion, flame, fire, venting of solid material, disassembly, cell leakage, or rupture of cell or battery within 24 hours after the completion of the test. The cell or battery shall not produce shrapnel within 24 hours after the completion of the test.

b. Dimensional distortion beyond specified limits.

c. Improper operation of safety devices that create an unsafe condition.

d. Cracking of cases or covers.

e. Mechanical failure of any part.

f. Liquid leakage or spilling of liquid at any time during the test. Liquid discharge visible from outside the battery or detectable within the battery shall not occur during testing to this specification, unless otherwise specified. Battery vent ports shall be open and rupture discs in place.

g. Breakdown of insulation, flashover, arcing, stripping of metal plating from any component part, corrosion of metal parts, or loosening of protective coating from the battery container or cover.

h. Deterioration of cell or battery identification markings.

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i. Charred or discolored areas of the cell or battery.

j. Opening of cell or battery vents.

k. Cracking, pitting, chipping, scaling, corrosion, or other deleterious effects on the external and internal metal or component parts of each cell or battery.

l. Cracking, peeling, fading or discoloration of the container and battery markings.

m. Improper indication by state of charge indicator or other state of health indicators.

3.8.2 Thermal protection. Thermal protection components shall be incorporated into the battery such that discharge and charge are inhibited, if the battery or cell operating temperatures are outside of the bounds specified by the battery manufacturer. Thermal protection shall be a functionality of the battery management system (BMS) as specified in 3.10.8 and table IX.

3.8.3 Potting. Cell or battery vents shall not be potted over. If potting is essential, care shall be exercised to ensure that venting shall not be obstructed. Potting shall not adversely affect cell or battery thermal management.

3.8.4 Pressurized batteries. Battery enclosure shall incorporate a safety venting device or be designed and manufactured so as to preclude a violent rupture as a result of cell venting. The design and construction shall not degrade the vent.

3.8.5 Interchangeability restraints. The battery shall be constructed such that the internal battery cells are not interchangeable, unless otherwise specified.

3.8.6 Shorting protection. The battery shall be delivered and stored with the leads or connector plugs protected against accidental shorting.

3.8.7 Age documentation. The manufacturer shall provide the following documentation with each delivery of cells or batteries:

a. The maximum age of cells or cells assembled into batteries, from the time of their initial manufacture to the time of their shipping date, shall be not greater than 180 days.

b. Batteries or cells shall be submitted for Government testing within 30 days of battery assembly.

3.8.8 General battery safety requirements. All lithium batteries shall be reviewed, tested and approved in accordance with S9310-AQ-SAF-010. Before listing on the qualified products list, the manufacturer shall comply with the requirements of S9310-AQ-SAF-010, unless modified by the applicable specification sheet (see 6.2.h, 6.18.1 and 6.18.2).

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3.8.9 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.8.10 Additional battery safety requirements. The penetration test (see 3.9.14) and Crush test (see 3.9.15) requirements specified herein are additional safety tests that shall inform the safety assessment testing performed in accordance with NAVSEA S9310-AQ-SAF-010. As such, the inability of the lithium battery to meet this requirement does not necessarily result in an automatic rejection of the battery for service use. Test units that fail to meet this requirement will be rejected only if a technical evaluation of the test results by the Naval Ordinance Safety and Security Activity (or another applicable authority) establishes that rejection is the appropriate course of action.

3.9 Detail requirements.

3.9.1 Condition of battery. The battery container shall not exhibit scratches, gouges, dents, torn material, leakage, or other damage when inspected in accordance with 4.5.1 (see 6.6).

3.9.2 Visual and mechanical. Batteries shall comply with 3.5, 3.7 and Table II, when subjected to the examination of 4.5.2.

3.9.3 Dimensions and weight. The dimensions and weight of batteries shall comply with the applicable specification sheet when subjected to the inspection of 4.5.3.

3.9.4 Dielectric strength and insulation resistance. The battery and its components shall meet the requirements of 3.8.1 after being tested in accordance with 4.5.5. The leakage current shall not be greater than 0.5 milliampere$_{RMS}$ (RMS - Root Mean Square) between each of the items listed in 4.5.5.a through 4.5.5.c. The insulation resistance shall be not less than 1 megohm between each of the items listed in 4.5.5.a through 4.5.5.c.

3.9.5 Vent tubes. If applicable, the vent tubes shall support the weight of twice the battery weight without shearing, breaking, bending, or deterioration at the point of connection with the battery or other distortion of the case after being tested in accordance with 4.5.6 (see 4.3.4). The battery shall also meet the requirements of 3.8.1.

3.9.6 Open circuit voltage. The cell or battery open circuit voltage shall be not greater than 29.0 volts before being tested in accordance with 4.5.4.

3.9.7 Capacity and electrical performance. The capacity (see 6.15.1 and 6.15.9) shall be as shown on the applicable specification sheet. The discharge requirements and associated parameters shall be as shown in table III (see 6.9). The battery shall produce not less than the cutoff voltage (see 6.15.3) at the designated discharge rate (see 6.15.2) for not less than the indicated length of time at the listed temperature after being tested in accordance with 4.5.8.

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Battery capacity shall vary not greater than ±5 percent within a given inspection lot. The tolerance of the discharge current shall be ±1percent of the specified value. The cell or battery temperature shall be not greater than 82 °C (180 °F) at any time during a capacity test. The state of charge indicator shall produce the proper indication corresponding to the battery capacity at the states of highest charge and lowest charge (i.e., at cut-off voltage) before and after the test of 4.5.8. Indications at intermediate levels shall be as specified in the applicable specification sheet.

3.9.8 High and low voltage charge test. The battery shall produce not less than 80 percent of its capacity when being tested in accordance with 4.5.9.c. The battery shall produce not less than 100 percent of its capacity when being tested in accordance with 4.5.9.g. At all times during the charge and discharge of 4.5.9 the battery shall conform to the requirements of 3.8.1.

3.9.9 Strength of receptacle (MIL-PRF-18148/3 or equivalent). The receptacle shall withstand a torsion force of 8.1 ±0.7 Newton-meters (6 ±0.5 foot-pounds) when being tested in accordance with 4.5.10. The battery shall also meet the requirements of 3.8.1.

3.9.10 Cycling. The battery shall successfully complete not less than 600 cycles or the number of cycles specified in the applicable specification sheet (see 6.15.4) when being tested in accordance with 4.5.11. Unless stated otherwise, the battery shall be capable of switching between the charge state and discharge state without any time delay for all tests shown in the verification section. The inability of the battery to meet the requirements of Table III, for the respective temperature requirement, for three consecutive discharges shall constitute a failure.

3.9.11 Functional discharge. The battery potential shall be not less than 20.0 volts at not less than 3 minutes into the discharge when being tested in accordance with 4.5.17, 4.5.24, 4.5.26, 4.5.27, 4.5.28 through 4.5.31.

3.9.12 Electromagnetic interference. The battery shall not exhibit any malfunction or degradation of performance when subjected to electromagnetic interference. The battery shall meet the applicable requirements of MIL-STD-461 and MIL-STD-464 when being tested in accordance with 4.5.12.

3.9.13 Temperature rise and float. The battery shall perform as follows when being tested in accordance with 4.5.15. If the charging current rises, its increase (in amperes) compared to the current's minimum value shall be not greater than 10 percent of the battery's rated capacity (for example, 1.5 amperes for a 15-ampere-hour battery) at any time during charge. The battery shall meet requirement (1) of table III and the requirements of 3.8.1.

3.9.14 Penetration test. Batteries shall not fail violently when their containers are penetrated by sharp objects. Samples tested in accordance with 4.5.36 shall not explode nor create shrapnel. When tested in accordance with 4.5.36.a, the temperature of each test sample shall be not greater than 170 °C (338 °F). For purposes of this abusive test and safety assessment,

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3.8.1 may be disregarded. See applicable specification sheet for the location of the test sample temperature measurement.

3.9.15 Crush test. When tested in accordance with 4.5.37, the cell or battery shall not explode nor create flame or fire within 24 hours after the completion of the test. The cell or battery shall not produce shrapnel within 24 hours after the completion of the test. For purposes of this abusive test and safety assessment, requirement 3.8.1 may be disregarded.

3.9.16 Ground storage. The cell or battery shall be capable of storage in uncontrolled temperature conditions from -40 °C to +50 °C (-40 °F to +122 °F) for not less than 30 days, after which it shall deliver not less than 50 percent of rated capacity after being tested in accordance with 4.5.34.

3.9.17 Shelf life. In order to maintain qualification, successful completion of the shelf life test shall be required. The cell or battery shall be capable of storage in uncontrolled temperature conditions from -40 °C to +50 °C (-40 °F to +122 °F) for not less than 18 months without damage (see 6.2.f). The capacity shall be recoverable when being tested in accordance with 4.5.35. The cell or battery shall also meet the requirements of 3.9.10 when being tested in accordance with 4.5.11.

3.9.18 Impact resistance (nonmetallic battery container only). The nonmetallic battery container shall not crack on the inside opposite the point of impact after being tested in accordance with 4.5.18.

3.9.19 Output performance. The battery shall produce not less than the potential specified in the applicable specification sheet during the test of 4.5.19.

3.9.20 Discharge and charge at 71 °C. The battery shall produce not less than the potential specified in the applicable specification sheet during the test of 4.5.20. When being discharged in accordance with 4.5.20.f, the battery shall show not less than 75 percent of its nominal rated capacity. The battery shall also meet the requirements of 3.8.1.

3.9.21 Cold and hot temperature charge. The battery shall produce not less than the capacity and potential specified in the applicable specification sheet after being discharged in accordance with 4.5.21. The battery shall also meet the requirements of 3.8.1.

3.9.22 Vibration. During the random vibration tests of 4.5.24, the fluctuation of the cell or battery voltage shall be not greater than 1.5 volts of the load voltage and shall not fall below the minimum closed circuit voltage specified in the applicable specification sheet. Also, the fluctuation of the cell or battery output current shall be not greater than 10 percent of the load current.

3.9.23 Transient drop test. Samples tested in accordance with 4.5.25 shall not show:

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a. Explosion, flame, fire, venting of solid material, disassembly, cell leakage, or rupture of cell or battery within 24 hours after the completion of the test.

b. The cell or battery shall not produce shrapnel within 24 hours after the completion of the test.

c. Liquid leakage or spilling of liquid at any time during the test. Liquid discharge visible from outside the battery or detectable within the battery shall not occur during testing.

3.9.24 Shock (crash safety). The battery shall perform as follows when being tested in accordance with 4.5.26 and 4.5.27.

a. The battery's normal attachment points (such as brackets, flanges, or hold-down bars) shall retain the entire battery intact in the test fixture during and after the test.

b. Mechanical bending and distortion are acceptable.

3.10 Battery management system (BMS). The manufacture shall provide a description of the BMS design (see 6.2.i).

3.10.1 Battery management system requirements. The battery shall contain an integrated BMS. The BMS shall not add volume or weight outside the limits specified in the battery specification sheet (see 3.1). See 4.5.38.1 for verification method.

3.10.2 BMS power. The BMS shall obtain its power from the following sources under the following conditions:

a. The BMS shall be powered by internal battery cells during the following conditions: when external power is not connected or during a deep discharge event (see 3.10.4). For a deep discharge event that causes the battery voltage to remain at 20.0 volts or less for a time exceeding 30 seconds, the BMS shall continue to be powered by internal battery cells. See 4.5.38.2.a for verification method of this requirement.

b. When external power is connected, the BMS shall be powered in parallel by the internal battery cells and external power source. See 4.5.38.2.b for verification method.

3.10.3 BMS electronics. When Battery Management System electronics are incorporated, the following requirements shall apply.

a. The battery and BMS battery electronics shall meet the requirements of MIL-STD-704. Guidance on testing for meeting these requirements for 28 VDC bus can be found in MIL-HDBK-704-8. See 4.5.38.3.a for verification method for MIL-STD-704 testing. Requirements for the individual type of batteries are specified in the applicable specification sheet (see 3.1).

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b. Battery BMS protective circuits, including battery disconnect function, shall be suitably protected from cell failure conditions within the battery such that the safety function is not compromised. See 4.5.38.3.b for verification method.

c. Battery BMS warning circuits, including fault and built in test (BIT) functions, shall be suitably protected from cell failure conditions with the battery system such that the warning signal is not compromised. See 4.5.38.3.c for verification method.

3.10.4 Deep discharge recovery. The protective circuitry of the battery shall inhibit discharge of the battery when the battery voltage is discharged down to 20.0 volts or less and following a time delay of 30 seconds. The time delay is provided due to voltage transients caused by engine starting, emergency, or fault clearing loads. If, at any time during the time delay, the battery voltage is greater than 20.0 volts (either under load or open circuit), then the time delay timer shall be reset. If the battery voltage is 20.0 volts or less during the whole duration of the time delay, then the battery discharge shall remain inhibited (no on-off-on-off chatter) until the battery is charged and pack voltage rises to 24.0 volts or greater. See 4.5.38.4 for verification method.

3.10.5 State of charge (estimated), state of health (estimated), and fault indicator. The battery shall have a digital display with a location defined in the applicable specification sheet and that will display the battery state of charge (SOC), state of health (SOH), and fault information in accordance with the following:

a. SOC Indicator. The SOC shall be calculated and stored by the BMS and shown on the digital display when the digital display push button (PB) (see 6.15.4) is activated. See 6.15.11 for definition of SOC. See 4.5.38.5.a for verification method. The manufacturer shall provide the Beginning of Life (BOL) capacity of the battery and a description of how the BMS software will estimate the SOC in accordance with the following formula:

(1). State of Charge = [Measured Capacity] / [Measured Full Charge Capacity] x 100%

b. SOH Indicator. The SOH shall be calculated and stored by the BMS and shown on the digital display when the digital display PB is activated. See 6.15.12 for the definition of SOH. See 4.5.38.5.b for verification method. The manufacturer shall provide a description of how the BMS software will estimate the SOH in accordance with the following formula:

(1). State of Health = [Measured Full Charge Capacity] / [BOL Capacity]

c. Fault indicators. Battery faults including Built-In-Test outcomes (see 3.10.6) shall be displayed when the digital display PB is activated in accordance with table IX. Fault specifications and the associated BMS responses are shown in table IX. The fault message reported on the digital display shall state a fault code and fault condition description. See 4.5.38.5.c for verification method.

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(1). Long-term fault data storage (non-volatile). Long-term data storage of battery fault data shall be stored in non-volatile memory in accordance with 3.10.14. Long-term fault data storage shall support the recording of the last five usage events (6.15.17) of the battery. See 4.5.38.5.b.(1) for verification method.

(2). Short-term fault data storage (Volatile). Short-term fault data shall be stored in volatile memory for each usage event (see 6.15.17) in accordance with 3.10.14. See 4.5.38.5.b.(2) for verification method.

3.10.6 Built-in test. The BMS shall perform built-in tests (BITs) to monitor and report the status of the fault conditions shown in table IX. If the BITs identifies a fault, then the BMS shall report that fault to the digital display when the PB is activated. BIT diagnostics shall be capable of isolating faults at the following ambiguity groups: cells, strings, pack, sensors, heater blankets, and BMS. BITs shall be executed automatically upon the start of a usage event and shall monitor for faults at a frequency of at least every 1-minute or faster until the end of the usage event. The BITs shall not degrade the performance of the battery. BIT diagnostics shall meet 95 percent or greater probability of detecting all battery faults shown in table IX. See 4.5.38.6 for verification method.

3.10.7 Voltage monitoring. The BMS shall be capable of monitoring the voltages associated with the following elements of the battery: cell, string, pack, and terminal. The battery shall meet this requirement after being tested in accordance with 4.5.38.7. The following requirements apply to monitoring of these voltages:

a. Battery cell, string, pack, and terminal voltages shall be monitored and stored in accordance with 3.10.13, table IV, and table V. See 4.5.38.7.a for verification method.

b. Cells at the same potential due to parallel electrical connections are permitted to be monitored and data stored as a single group. See 4.5.38.7.b for verification method.

c. The sample rate of cell, string, pack, and terminal voltages shall be in accordance with 3.10.6. See 4.5.38.7.c for verification method.

d. Manufacturers shall report the upper and lower voltage limits that apply to the cell, string, pack, and terminal elements. The upper and lower voltage limits shall be stored in non-volatile memory and referenced by the BMS for determining a fault related to voltage. See 4.5.38.7.d for verification method.

e. The BMS shall detect and respond to under-voltage and over-voltage faults in accordance with table IX. See 4.5.38.7.e for verification method.

3.10.8 Temperature monitoring. The BMS cell temperature monitoring, battery temperature monitoring, and capability shall be as follows:

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a. Manufacturer shall report the number of cell and battery temperature sensors required for safe operation of battery during qualification and conformance inspection (see 4.2 and 4.3) and the battery's service life. See 4.5.38.8.a for verification method.

b. Cell and battery sensor readings shall be monitored and stored in accordance with 3.10.13, table IV, and table V.

c. Manufacturer shall provide the minimum and maximum temperature limits for safe operation of the cells used in the battery during charge and discharge.

d. Temperature limits shall be stored in the non-volatile software memory and be used to indicate temperature faults. See 4.5.38.8.d for verification method

e. If a cell or string of cells have exceeded temperature limits, then the BMS shall inhibit the charge of the string to avoid an unsafe condition in accordance with table IX. See 4.5.38.8.e for verification method.

f. Any temperature fault shall displayed on the battery display when battery display push button is pushed in accordance with table IX. See 4.5.38.8.f for verification method.

3.10.9 Current monitoring. The BMS battery current monitoring, string current monitoring, and charge and discharge current inhibit capability shall be as follows:

a. Battery string and pack current shall be monitored and stored in accordance with 3.10.13, table IV, and table V. See 4.5.38.9.a for verification method.

b. The maximum charge and discharge current that the battery cell chemistry and battery pack design can be subjected to shall be reported by the battery manufacturer and shall be capable of satisfying the emergency, engine start, and fault clearing load requirements shown in the applicable specification sheet. The limits shall be stored in non-volatile memory. See 4.5.38.9.b. for verification method.

c. The BMS shall detect and respond to an over-current on discharge or charge based on limits required in 3.10.9.b and in accordance with table IX.

d. If a cell or string of cells exceeds the charge and discharge current limits, then the BMS shall inhibit the charge and discharge of that string of cells to avoid an unsafe condition. See 4.5.38.9.d for verification method.

3.10.10 Balance monitoring. The BMS shall provide balancing of the cells as follows:

a. The battery shall perform balancing of cells as appropriate to maintain maximum battery capacity and life. See 4.5.38.10.a for verification method.

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b. The BMS shall be capable of balancing the cells during the first hour of flight in accordance with the balancing load profile in the applicable specification sheet. If a balancing profile is unavailable, then the BMS shall be capable of balancing the cells during the test of 4.4.2. See 4.5.38.10.b for verification method.

c. The BMS shall monitor and store cell balancing, string balancing or module balancing, and battery capacity in accordance with 3.10.12, table IV, and table V. See 4.5.38.10.c for verification method.

d. Balancing may be terminated when the difference between the maximum and minimum cell voltage is less than 50 mV +/- 5 mV. See 4.5.38.10.d for verification method.

e. The BMS shall detect and respond to balancing faults in accordance with table IX. See 4.5.38.10.e for verification method.

f. Cell balancing shall take into account cell temperature to avoid overcharge of the cell. See 4.5.38.10.f for verification method.

3.10.11 Short circuit and in-rush current protection. Battery and BMS short circuit protection shall be as follows:

a. The BMS shall protect the battery from an external short circuit of 0.02 ohms.

b. If the short circuit protection is by-passed, the battery shall not fail in an unsafe manner when subjected to an external short circuit of 0.02 ohm. See 4.5.38.11.b for verification method.

c. The BMS shall protect the battery from excessive in-rush current that may damage the charging path within the battery. See 4.5.38.11.c for verification method.

d. The BMS shall detect and respond to external short-circuit in accordance with 3.10.12.e and table IX. See 4.5.38.11.d for verification method.

e. The BMS shall detect and respond to internal short-circuit in accordance with 3.10.12.e and table IX. See 4.5.38.11.e.

3.10.12 Data storage. The following requirements apply to the data storage of the battery:

a. Frequency. The battery parameters specified in table IV shall be stored in memory during each usage event (see 6.15.17). If the battery is operating without any active faults, then the battery parameters specified in table IV shall be stored in memory at a frequency of every one minute or faster. If the BMS detects a fault in accordance with table IX during a usage event (see 6.15.17), then the battery parameters specified in table IV shall be stored in memory at a frequency of every 10 seconds or faster until the fault becomes inactive. Additionally, the battery parameters specified in table IV shall be stored in memory at a frequency of every 10

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seconds or faster during the 30-second time delay of a deep discharge event (see 3.10.4). See 4.5.38.12.a for verification method.

b. Storage Format. The BMS shall store either non-formatted (binary) or formatted data related to battery performance and safety as specified in 3.10.5 through 3.10.14. A non-proprietary PC-based tool shall be used to retrieve and format the data (if applicable) such that is made available on the PC as a comma separated value (CSV) file, referred to herein as a usage event file (UEF). The UEF shall contain performance and safety data from at least the last five (5) usage events (see 6.15.17). Data recording, either non-formatted or formatted, shall occur within at least 1 second of the start of a usage event and occurs at a frequency specified in 3.10.12.a. See 4.5.38.12.b for verification method. The following requirements apply to the format of the UEF:

(1) Data fields. One data field (i.e., comma separated header) shall be created for each parameter specified in table IV.

(2) Data input. Data for each parameter shown in table IV shall be written to the UEF and associated with a date and time stamp in accordance with table V. Each data and time stamp represents a new row of data input to the UEF and shall be generated a frequency specified in 3.10.12.a.

(3) Data storage location. The data shall be stored in either volatile or non-volatile memory, such that the UEF contains performance and safety data from at least the last five (5) usage events (see 6.15.17).

c. Software and Fault Limits. Software and Fault Limits shall be stored in non-volatile memory. See 4.5.38.12.c for verification method.

d. Historical performance data. The BMS shall update, calculate, and store in non-volatile memory the following data after each usage event. See 4.5.38.12.d for verification method. The Historical Data shall be stored in a separate data file, distinct from other data files, labeled as Historical Data File (HDF).

i. Minimum and maximum temperature of the battery and strings
ii. Minimum and maximum voltage of the battery and strings
iii. Maximum discharge current
iv. Maximum charge current
v. Cumulative number of charge cycles completed
vi. Cumulative ampere-hours charged
vii. Cumulative ampere-hours discharged

e. Historical fault data. The BMS shall update and store historical fault data in a separate data file from other data files, labeled as Historical Fault File (HFF). The HFF shall update after

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each usage event and shall record the time, date, fault name, fault code, fault description message, and fault condition as described in table IX. See 4.5.38.12.e for verification method.

f. Storage capacity. The BMS shall store at least 7.0 hours of data per usage event (see 6.15.17) at a frequency in accordance with 3.10.12.a. The data sets associated with the five (5) most recent usage event shall be stored in non-volatile memory and be accessible by external communications (see 3.10.13) with the battery. If a usage event exceeds 7.0 hours, then the BMS may overwrite or delete the earliest data points such that memory is available for the latest data points and the BMS maintains 7.0 hours of data per usage event.

3.10.13 External communications. External communications is to allow for data stored by the BMS to be analyzed by maintenance shop equipment and shall be as follows:

a. Data stored (3.10.12) in the BMS memory shall be capable of transmitting to an external portable computer by the communication protocol shown in the applicable specification sheet. If the specification sheet does not list a required communication protocol, then RS-485 shall be used for transmitting data. See 4.5.38.13.a for verification method.

b. External communication shall be accomplished through the sensor receptacle J2. See 4.5.38.13.b for verification method.

3.10.14 BMS enhancement capability. The BMS bandwidth, volatile memory and non-volatile memory shall be sized at least 50 percent greater than what is required by MIL-PRF-29595 and the battery specification sheet (see 3.1) for software code, limit data, BIT, fault data, and data storage.

3.10.15 Maintenance equipment compatibility. The following requirements shall be met to ensure proper maintenance of the battery:

a. Battery manufacturer provides recommended charge and discharge rates and times to perform proper maintenance on Battery Charger/Analyzer, P/N: MSD-970-1, RF80-K, and RF80-M.

b. Battery shall be compatible with Battery Charger/Analyzer P/N: MSD-970-1, RF80-K and RF80-M. See 4.5.38.15.a, b, c, d for verification method.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

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4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in this general specification (see 3.1, 6.4, 6.5, 6.18.1, 6.18.2, and Table VI), the applicable specification sheet, and Chapter 13 of S9310-AQ-SAF-010 safety testing. The specification sheets also list the order in which the tests shall be conducted. The samples shall be representative of the items intended to be supplied under this specification. The samples shall not be produced with the use of any equipment or procedure not normally used in the production of the battery. If the battery has an integral charger, then all qualification samples shall contain the integral charger; in addition, one charger shall be supplied separately from the battery. If receptacles that are dimensionally equivalent to MIL-PRF-18148/3 receptacle are used, then five worm gear retaining pins shall be furnished for inspection. All samples shall be marked with identifying information in accordance with 3.7 (see 6.2.d).

4.3 Conformance inspection. Conformance inspection shall consist of the tests specified in the applicable specification sheet (see 3.1, 6.6, 6.7, 6.8 and table VI). Batteries produced under this specification shall be identical in every respect to the qualification sample tested and found satisfactory, except for changes previously approved by the Government. Any unapproved changes from the qualification sample shall constitute cause for rejection.

4.3.1 Inspection of product. Every battery shall be subjected to and pass group A tests; each shall also have been part of the same inspection lots from which group B samples were selected and which passed group B testing. Batteries produced under this specification shall successfully complete groups A and B inspection (see 4.3.4 and 4.3.5).

4.3.2 Inspection lots. Units shall be compiled into inspection lots (see 6.15.7 and 6.15.13). Unless otherwise specified by the contract or purchase order, the inspection lot size shall be determined as follows:

a. When the entire stock quantity specified by a contract, delivery order, or purchase order is to be delivered at one specified time, no more than two lot codes (see 3.7.6) shall appear among the units in a particular inspection lot. A specific lot code shall not be found in any more than two inspection lots.

b. When the entire contract quantity is to be delivered in several shipments over a period of months or years (such as direct vendor delivery style contracts), the inspection lot size and resulting sample quantity shall consist of the quantity of batteries that the Government or contractor estimates will be delivered during a period of up to three months.

4.3.3 Sample size and selection for groups A and B inspection. Each unit shall undergo group A inspection (see 6.8). Unless otherwise specified by the contract or purchase order, the sample size for group B shall be as shown in table VII (see 6.2.d).

4.3.4 Group A inspection. Each unit shall be subjected to the examination and test requirements in the applicable specification sheet (see 6.8). Each unit shall meet the requirements for group A inspection in the applicable specification sheet. Defective units

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discovered during group A inspection shall be individually rejected. Corrective action for group A strength of vent tube failures (see 3.9.5 and 4.5.6) shall be provided to the qualifying activity for approval before the start of manufacture of the next production lot.

4.3.5 Group B inspection. The group B tests shall be performed in accordance with the applicable specification sheet (see 6.8).

4.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the following test conditions.

4.4.1 Temperature and storage conditions. Unless otherwise specified in the description of the test, all measurements, tests, and capacity discharges shall be made at room temperature, ambient atmospheric pressure, and ambient relative humidity. Unless otherwise specified in the description of the test, room temperature for all tests except group A inspection shall be 25° ±5 °C (77 °F ±9 °F). Room temperature for group A inspection shall be 24 °C ±8 °C (75°±15 °F). Unless otherwise specified in the description of the test, all high and low temperature charge and discharge tests shall be conducted at the temperature indicated in the applicable test paragraph; a tolerance of ±2 °C (4 °F) shall be allowed. Stabilize the battery in the environmental chamber at the specified temperature for 22 ±2 hours before discharge.

4.4.2 Constant potential charging method. Unless otherwise specified in the applicable specification sheet, constant potential charge the batteries at 28.25 ±0.25 volts for 2.0±0.1hours. The power source used for charging shall have a current capacity of not less than 300 amps to simulate an aircraft 28 volt DC bus. If the battery contains a heater circuit, the circuit shall be energized during the complete charge (see 3.5.9).

4.4.3 Discharges. For all discharges, the current, resistance, or voltage (whichever is applicable) shall be maintained within ±1 percent of the value specified in the applicable test at all times, and the duration of the discharges shall be within ±2 percent of the nominal discharge time. Unless otherwise specified by the test, all discharges shall be continuous.

4.4.4 Mounting. For the vibration and mechanical shock tests, the mounting shall be by normal aircraft attachment points (see figure 3). Vibration isolators shall not be used. For all other tests, the bottom of the battery shall be supported on a rigid surface.

4.4.5 Repeat upon failure of test. Unless otherwise specified in the description of the test, if the battery fails to meet the specified minimum requirements of any electrical test except 4.5.8, 4.5.9, 4.5.19, or 4.5.21, the sample shall have failed the test. If the battery fails 4.5.8, 4.5.9, 4.5.19, or 4.5.21, the battery shall be recharged as described in 4.4.2, after which the test will be repeated. Inability to meet the specified minimum requirements during the repeated test shall constitute a failure of that particular sample. If the battery meets the requirements of the repeated test, the specified test schedule shall be resumed.

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4.5 Methods of inspection. The tests on each battery shall be performed in the order listed in the applicable specification sheet. The following tests and examinations (see 6.11) shall be performed as follows.

4.5.1 Incoming inspection. Examine the battery container for the requirements of 3.9.1.

4.5.2 Visual and mechanical examination. Examine the battery for the requirements of 3.9.2.

4.5.3 Dimensions and weight. Examine the battery for the requirements of 3.9.3 and 3.10.1. Measure the weight of each sample. Examine each dimension of the following samples:

a. Qualification inspection: Each sample.

b. Group A inspection: On 2 percent of the batteries in the inspective lot or not less than one battery from the inspection lot (whichever is greater).

c. Group B inspection: On one battery from the lot.

4.5.4 Open circuit voltage. Measure and record the open circuit voltage of the test sample before and after each test. Examine the items for the requirements of 3.9.6.

4.5.5 Dielectric strength and insulation resistance test. During group A inspection, perform this test on 5 percent of the units. For dielectric strength, apply a DC potential of 500 ±25 volts for 60 +5, -0 seconds to the following combination of components. For insulation resistance, apply an alternating 500 ±25 $volts_{RMS}$ for 60 +5, -0 seconds to components listed below. Test the following combinations of components:

a. The current-carrying parts of the battery and the battery case.

b. The heater pins of the auxiliary connector and the current-carrying parts of the battery.

c. The pins of the auxiliary connector and the battery case. Measure the insulation resistance of the components listed above in accordance with MIL-STD-202-302, test condition B, for 60 seconds. Examine the items for the requirements of 3.9.4.

4.5.6 Strength of vent tubes test. If vent tubes are applicable, mount a weight equivalent to twice the battery weight on top of the test sample battery. Lift the test sample-battery by both vent tubes to a height of 12.7 centimeters (5 inches) (nominal) above the work surface for a period of 10 seconds. Observe any evidence of shearing, binding, or deterioration at the point of connection of the vent tube and the test sample battery. Examine the test sample battery for the requirements of 3.9.5. During group A inspection, perform this test on 5 percent of the units. If a failure occurs during group A inspection, then 100 percent of the units shall be tested in that lot (see 4.3.4).

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4.5.7 Electrical test preparation. Discharge the battery at the 1.0 C-rate to 20.0 volts. Charge the battery at 28.25 ±0.25 volts for 2.0 ± 0.1 hours.

4.5.8 Capacity discharge. Stabilize the battery at the temperature(s) specified by the applicable specification sheet (e.g., -40 °C, 20 °C, and 55 °C). Discharge the battery at the 1.0 C-rate to the cutoff voltage specified by the applicable specification sheet. Constant potential charge the battery in accordance with 4.4.2. For group A conformance inspection, any charge method (4.4.2 or other method not listed in section 4) may be selected as long as all batteries are charged by the same method and still meet all the requirements of this specification. Examine the battery for the requirements of 3.9.7 and requirement (1) of Table III. For group A, requirement (4) of Table III shall be used.

4.5.9 High and low voltage charge test. This test consists of the following steps.

a. Discharge the battery at the C-rate to a cutoff voltage of 20.0 volts.

b. Constant potential charge the battery in accordance with the applicable specification sheet to 80 percent SOC.

c. Discharge the battery at the C-rate to a cutoff voltage of 20.0V.

d. Constant potential charge the battery at 28.25 ±0.25 volts for 16.0 ±0.1 hours.

e. Discharge the battery at the C-rate to a cutoff voltage of 20.0V.

f. Constant potential charge the battery in accordance with the applicable specification sheet to 100% SOC.

g. Discharge the battery at the C-rate to a cutoff voltage of 20.0V.

h. Examine the battery for the requirements of 3.8.1 and 3.9.8.

4.5.10 Strength of receptacle test. If the battery uses a MIL-PRF-18148/3 receptacle or equivalent, insert a grooved steel shaft that mates with the receptacle socket such that the socket pins are secured in the steel shaft. Subject the steel shaft to a torsion force of 8.1 ±0.7 Newton-meters (6 ±0.5 foot-pounds). Examine the receptacle mounting for the requirements of 3.9.9.

4.5.11 Cycling test. Subject the battery to 600 cycles of discharge-charge-open circuit and to the temperature environments below. This test consists of the following steps.

a. Constant potential charge the battery at 28.25 ±0.25 volts for 2.0 ±0.1 hours at room temperature (see 4.4.1).

b. First sequence: Cold temperature environment (cycles 1 through 100).

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(1) Stabilize the battery temperature at -18 °C (0° F) (see 4.4.1). Perform steps 4.5.11.b(2) through 4.5.11.b(5) with the battery in the -18 °C (0 °F) temperature environment.

(2) Discharge the battery at the 1.0 C-rate for 60.0 + 0.0 / - 0.1 minutes or to cutoff voltage of 20.0 volts

(3) Immediately after completing step 4.5.11.b(2), constant potential charge the battery at 28.25 ±0.25 volts for 2.0 ±0.1 hours.

(4) Place the battery on open circuit for a period of 1.0 ±0.1 hour.

(5) Repeat above steps 4.5.11.b(2) through 4.5.11.b(4) for a total of 100 cycles.

c. Second sequence: Hot temperature environment (cycles 101 through 200).

(1) Stabilize the battery temperature at 43 °C (110 °F) (see 4.4.1). Perform steps 4.5.11.c(2) through 4.5.11.c(5) with the battery in the 43 °C (110 °F) temperature environment.

(2) Discharge the battery at the 1.0 C-rate for 60.0 + 0.0 / - 0.1 minutes or to cutoff voltage of 20.0 volts

(3) Immediately after completing step 4.5.11.c(2), constant potential charge the battery at 28.25 ±0.25 volts for 2.0 ±0.1 hours.

(4) Place the battery on open circuit for a period of 1.0 ±0.1 hour.

(5) Repeat above steps 4.5.11.c(2) through 4.5.11.c(4) for a total of 100 cycles.

d. Third sequence: Ambient temperature environment (cycles 201 through 300).

(1) Stabilize the battery temperature at 24 °C (75 °F) (see 4.4.1). Perform steps 4.5.11.d(2) through 4.5.11.d(3) with the battery in the 24 °C (75 °F) temperature environment.

(2) Discharge the battery at the 1.0 C-rate for 60.0 + 0.0 / - 0.1 minutes or to cutoff voltage of 20.0 volts

(3) Immediately after completing step 4.5.11.d(2), constant potential charge the battery at 28.25 ±0.25 volts for 2.0 ±0.1 hours.

(4) Place the battery on open circuit for a period of 1.0 ±0.1 hour.

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(5) Repeat above steps 4.5.11.d(2) through 4.5.11.d(4) for a total of 100 cycles.

e. Repeat steps 4.5.11.b through 4.5.11.d for a total of 600 cycles.

f. Stabilize the battery temperature at 24 °C (75 °F) and discharge at the 1C-rate to 20.0 volts.

g. Examine the battery for the requirements of 3.9.10.

4.5.12 Electromagnetic interference. Electromagnetic interference shall be conducted in accordance with MIL-STD-461 test procedures of CE102, CS101, CS114, CS115, CS116, RE102 and RS103 (30 MHz to 18 GHz at 200 v/m). The test setup shall include the battery, a battery charger (power source), and DC load all connected to a common bus. Line impedance stabilization networks (LISNs) shall be installed on both the 28 VDC power leads and on the 115 VAC, single-phase heater circuit. The charger and DC load shall be located outside of the EMI chamber. Testing shall be conducted for two modes of operation, discharging at a 5-ampere rate with the charger off and charging/floating on the load bus at any rate of charge with the charger on. The battery shall remain on and connected to the load bus throughout the test and shall not disconnect from the bus. The functionality of the battery shall be demonstrated and recorded before and after each EMI test by discharging the battery to cutoff and recharging the battery. Examine the battery for the requirements of 3.10.5, 3.10.6, and 3.9.12.

4.5.13 Charge and discharge test at low temperature. This test consists of the following steps.

a. Constant potential charge the battery in accordance with 4.4.2 at room temperature (see 4.4.1).

b. Place the battery in a temperature chamber at -18 °C (0 °F) and stabilize the battery (see 4.4.1). Perform the following with the battery in the temperature environment.

c. Discharge the battery at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

d. Examine the battery for the requirements of 3.9.7 and requirement (3) of Table III.

e. Constant potential charge the battery in accordance with 4.4.2.

f. Immediately following charge, remove the battery from the test chamber. Discharge the battery at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

g. Examine the battery for the requirements of 3.9.7 and requirement (3) of Table III.

4.5.14 Discharge while inverted. This test consists of the following steps.

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a. Constant potential charge the battery in accordance with 4.4.2 at room temperature (see 4.4.1).

b. Invert the battery (see 6.15.10).

c. Discharge the battery at the 4.0C-rate for 5 minutes or to a cutoff voltage of 20.0 volts, whichever occurs first. At 2.5 minutes into the discharge, place the battery upright.

d. Examine the battery for the requirements of 3.6.2 and requirement (4) of Table III.

4.5.15 Temperature rise and float test. This test consists of the following steps.

a. Constant potential charge the battery in accordance with 4.4.2.

b. Place the battery in a temperature chamber at 49 °C (120 °F) for not less than 12 hours.

c. Perform the following with the battery in the 49 °C (120 °F) temperature environment.

d. Discharge the battery at the 4.0C-rate for 5 minutes or to a cutoff voltage of 20.0 volts, whichever occurs first.

e. Immediately after the 4.0C-rate discharge and with the battery still in the temperature chamber, constant potential charge the battery at 28.6 ±0.10 volts for 24 ±0.1 hours.

f. Stabilize the battery temperature at room temperature (see 4.4.1). Discharge the battery at the 1.0 C-rate to a cut-off voltage of 20.0 volts.

g. Record voltage, current, and time throughout the test.

h. Repeat steps 4.5.15.a through 4.5.15.f for a total of 3 cycles.

i. Remove the battery from the temperature chamber.

j. Examine the battery for the requirements of 3.9.13.

4.5.16 Thermal protection. Place each test sample in a temperature chamber at 82 °C ±2 °C (180 ±4 °F) for not less than 2 hours. Check each sample to verify that the thermal protection is closed. Raise the temperature to 88 °C ±2 °C (190 °F ±4 °F). After 45 minutes, check each sample to verify that the switch or thermal protection is open. Lower the temperature to 82 °C ±2 °C (180 °F ±4 °F) and check each sample after 45 minutes to verify that the thermal protection is closed. Examine the thermal protection for the requirements of 3.8.2.

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4.5.17 Physical integrity at high temperature test. This test consists of the following steps.

a. Constant potential charge the battery in accordance with 4.4.2.

b. Discharge the battery at the 1.0C-rate to a cutoff voltage of 20.0 volts.

c. Constant potential charge the battery in accordance with 4.4.2.

d. Place the battery in a temperature chamber at 85 °C (185 °F) for 16 ±0.1 hours.

e. Remove the battery from the temperature chamber. Lift the battery by its vent tubes.

f. Expose the battery to room temperature for 1 hour (see 4.4.1).

g. Discharge the battery at the 4.0C-rate for 3 minutes.

h. Examine the battery for the requirements of 3.9.11.

i. Stabilize the battery at room ambient conditions (see 4.4.1).

j. Constant potential charge the battery in accordance with 4.4.2.

k. Discharge the battery at the 1.0C-rate to a cutoff voltage of 20.0 volts.

l. Examine the battery for the requirements of 3.8.1 and requirement (1) of table III.

4.5.18 Impact resistance test (nonmetallic battery container). This test consists of the following steps.

a. Rest an undamaged nonmetallic container for not less than 24 hours after manufacture.

b. Place the container on a flat steel plate which is not less than 25 centimeters (1 inch) longer and wider than the container. Position the container such that the ball will strike one-third down from the top of the container and near the centerline of the sides.

c. Condition the container for 4 hours at 71 °C ±3 °C (160 °F ±5 °F).

d. Drop a free-falling 0.907 ± 0.023-kilogram (2.00 ±0.05-pound) solid steel ball one time on the container from a height of 2.29 meters (90 inches) (nominal).

e. Examine the container for the requirements of 3.9.18.

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f. Repeat step 4.5.18.c but use a temperature of -18 °C ±1 °C (0 °F ±2 °F). Repeat step 4.5.18.d but use a height of 1.52 meters (60 inches) (nominal). Repeat step 4.5.18.e.

g. Repeat step 4.5.18.c but use a temperature of -40 °C ± 1 °C (-40 °F ±2 °F). Repeat step 4.5.18.d but use a height of 1.27 meters (50 inches) (nominal). Repeat step 4.5.18.e.

4.5.19 Output performance test. This test consists of the following steps:

a. Constant potential charge the battery in accordance with 4.4.2

b. Stabilize the battery at a temperature of 65 °C (149 °F) for 22 ±2 hours on open circuit. Perform steps (c) through (e) with the battery in the 65 °C (149 °F) temperature environment.

c. Discharge the battery at the 1.0 C-Rate for 1.0 ±0.1 hour of 20.0 volts whichever occurs first.

d. Immediately after discharge, constant potential charge the battery in accordance with 4.4.2. Rest battery on open circuit for 22 ±2 hours.

e. Repeat steps (c) and (d) for a total of 5 cycles.

f. At the completion of cycle 5, stabilize the battery at a temperature of -29 °C (-20 °F) for 22 ±2 hours on open circuit (see 4.4.1).

g. Discharge the battery in accordance with applicable specification sheet to a cutoff voltage of 20.0 volts.

h. Examine the battery for the requirements of 3.9.19.

4.5.20 Discharge and charge at 71 °C. This test consists of the following steps.

a. Constant potential charge the battery in accordance with 4.4.2.

b. Stabilize the battery's temperature at 71 °C (160 °F) for 24 ±0.1 hours.

c. With the battery still in the temperature chamber at 71 °C (160 °F), discharge the battery in accordance with the applicable specification sheet.

d. With the battery still in the temperature chamber at 71 °C (160 °F), constant potential charge the battery in accordance with 4.4.2.

e. Stabilize the battery at room ambient conditions (see 4.4.1).

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f. With the battery at room temperature, discharge the battery at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

g. With the battery at room temperature, constant potential charge the battery in accordance with 4.4.2.

h. Examine the battery for the requirements of 3.9.20.

4.5.21 Cold and hot temperature charge test. This test consists of the following steps.

a. Constant potential charge the battery in accordance with 4.4.2.

b. Stabilize the battery's temperature at the test temperature specified in the applicable specification sheet (see 4.4.1). Perform steps 4.5.21.c through 4.5.21.e with the battery in the environmental chamber at the specified temperature.

c. Discharge the battery in accordance with the specification sheet.

d. Stabilize the battery's temperature at the test temperature specified in the applicable specification sheet.

e. Constant potential charge the battery in accordance with 4.4.2.

f. Immediately after completing 4.5.21.e, place the battery in a room temperature environment (see 4.4.1) and discharge the battery in accordance with the specification sheet.

g. Examine the battery for the requirements of 3.9.21.

4.5.22 Worm gear retaining pin hardness. Mount and polish five samples of the worm gear retaining pins, making polished metallurgical cross-sections. Use a 100-gram load and a Knoop micro-hardness tester to make 6 indentions on each of the five pins. Average the results for each pin. Calculate the hardness of the pins and convert the results to the Rockwell C scale. Examine the pins for the requirements of 3.5.5.

4.5.23 Low pressure (altitude). This test consists of the following steps:

a. Subject the test sample to explosive decompression in accordance with MIL-STD-810, method 500.4, procedure IV.

b. Perform the tests of 4.5.2 and 4.5.3.

c. Examine the battery for the requirements of 3.9.3.

d. Perform the tests of 4.5.4 and 4.5.8.

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e. Examine the battery for the requirements of 3.8.1, 3.10.5, and 3.10.6.

4.5.24 Vibration test. This test consists of the following steps:

a. Constant potential charge the battery in accordance with 4.4.2.

b. For all batteries using the MIL-PRF-18148/3 main battery receptacle: Connect two AN-6 cables to the positive terminal and another two AN-6 cables to the negative terminal. Clamp the cables to the test fixture at a cable length of 46 ±5 centimeters from the battery.

c. Mount the battery assembly, including connector and cables, to the test fixture via the normal battery attachment points (see 4.4.4).

d. Vibrate the battery along each orthogonal axis (in the axis order shown on figure 3) in accordance with the test levels of Table VIII to provide a composite level of 59.62 meters per second squared (6.08 g_{RMS}) (g_{RMS} – root-mean-square acceleration). The random vibration cycling time in each axis shall be 1 hour.

e. Constant potential charge the battery in accordance with 4.4.2 at 28.25 ± 0.50 volts during vibration. Discharge the battery at the 1.0 C-rate for 10 minutes before beginning each axis.

f. After completion of the vibration test, constant potential charge the battery in accordance with 4.4.2.

g. Discharge the battery at the 4.0C-rate for 3 minutes.

h. Examine the battery for the requirements of 3.8.1, 3.9.3, 3.10.5, 3.10.6, 3.9.11, and 3.9.22.

4.5.25 Transient drop test. This test consists of the following steps:

a. Constant potential charge the battery or cell in accordance with 4.4.2.

b. Perform four drops from a height of 122 centimeters (48 inches) allowing the test unit to bounce and naturally come to a resting position. After each drop visually inspect the test item and note any obvious damage. Document the impact points. The test item shall not be dropped subsequent times on the same face.

(1) Prismatic – Drop the test unit twice with a face parallel to the concrete surface and facing downward upon release, but need not be parallel upon impact. Drop the test unit twice with a corner parallel to the concrete surface and facing downward upon release, but need not be parallel upon impact.

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(2) Cylindrical – Drop the test unit twice with the axis of the cylinder parallel to the concrete surface and facing downward, but need not be parallel upon impact. Drop the test unit twice with the axis of the cylinder perpendicular to the concrete surface and facing downward, but need not be parallel upon impact.

c. Perform the tests of 4.5.2 and 4.5.3.

d. Examine the battery for the requirement of 3.9.23.

e. Discharge the battery at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

f. Examine the battery for the requirement of 3.9.23, 3.10.5, and 3.10.6.

4.5.26 Shock test (basic design). This test consists of the following steps:

a. Constant potential charge the battery in accordance with 4.4.2.

b. Mount the battery to the test fixture in accordance with 4.4.4.

c. Discharge the battery at the C/2 rate during the test.

d. Subject the battery to the shock pulse of figure 4, where P equals 20 *g* and D equals 11 milliseconds. Shock the battery in a normal upright position (see 6.15.10) and in two additional mutually perpendicular directions from the vertical. However, the battery shall not be mounted or shocked in the inverted position. Deliver three shocks in both directions of each orthogonal axis for a total of 15 impacts.

e. After each shock, examine the battery for the requirements of 3.8.1.

f. After the fifteenth impact, discharge the battery at the 4.0 C-rate for 3 minutes.

g. Examine the battery for the requirements of 3.10.5, 3.10.6, and 3.9.11.

4.5.27 Shock test (crash safety). This test consists of the following steps:

a. Constant potential charge the battery in accordance with 4.4.2.

b. Mount the battery to the test fixture in accordance with 4.4.4.

c. Subject the battery to the shock pulse of figure 4, where P equals 40 *g* and D equals 11 milliseconds. Shock the battery in a normal upright position (see 6.15.10) and in two additional mutually perpendicular directions from the vertical. However, the battery shall not be mounted or shocked in the inverted position. Deliver two shocks in both directions of each orthogonal axis for a total of 10 impacts.

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d. After each axis of shock, dismount the battery from the test fixture.

e. Examine the battery and its attachment points for the requirements of 3.9.24.

f. After the last impact, remove the battery from the test fixture. Discharge the battery at the 4.0 C rate for three minutes.

g. Examine the battery for the requirements of 3.9.11 and 3.9.24.

4.5.28 Altitude test. This test consists of the following steps:

a. Constant potential charge the battery in accordance with 4.4.1 and 4.4.2.

b. Immediately after completion of charge, place the battery inside an environmental chamber.

c. Within 15 minutes after completing step 4.5.28.b, lower the chamber temperature to -29 °C (-20 °F) and lower the pressure inside the chamber to simulate the ambient pressure at 18,288 meters (60,000 feet) altitude. Perform the remainder of the test with the battery in the temperature environment. Invert the battery for 2 minutes while the chamber air pressure is being reduced and then return the battery to its normal upright position (see 6.15.10).

d. Discharge the battery at the 4.0C-rate for 3 minutes. Examine the battery for the requirements of 3.9.11.

e. Constant potential charge the battery in accordance with 4.4.2 for 2.0 ±0.1 hours. Invert the battery for 5 minutes during the 2-hour charge and then return the battery to its normal upright position.

f. Open-circuit the battery and, within 5 minutes, return the chamber to ground ambient air pressure (see 4.4.1). Invert the battery for 2 minutes while the chamber air pressure is being increased to ground ambient and then return the battery to its normal upright position.

g. Discharge the battery at the 4.0C-rate for 3 minutes. Examine the battery for the requirements of 3.9.11.

h. Repeat step 4.5.28.c. Repeat steps 4.5.28.e, 4.5.28.f, and 4.5.28.g.

i. Examine the battery for the requirements of 3.6.2, 3.8.1, 3.10.5, and 3.10.6.

4.5.29 Humidity test. This test consists of the following steps:

a. Constant potential charge the battery in accordance with 4.4.2.

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b. Gradually raise the internal chamber temperature to 60 °C (140 °F) and the relative humidity to 95 ± 5 percent over a period of 2.0 ±0.1 hours.

c. Maintain the conditions of step 4.5.29.b for not less than 6 hours.

d. Maintain not less than 85 percent relative humidity and reduce the internal chamber temperature in 8.0 ±0.1 hours to 30 °C (86 °F) and 95 ±5 percent relative humidity.

e. Maintain the 30 °C (86 °F) and 95 ±5 percent relative humidity for an additional 8.0 ± 0.1 hours.

f. Repeat steps 4.5.29.b, 4.5.29.c, and 4.5.29.d for a total of 10 cycles (not less than 240 hours).

g. While the battery is still in the chamber, C-rate discharge the battery for 15 ±1 minute.

h. While the battery is still in the chamber, constant potential charge the battery in accordance with 4.4.2.

i. Remove the battery from the chamber and open-circuit the battery for 8.5 ±7.5 hours.

j. Examine the battery for the requirements of 3.8.1.

k. Discharge the battery at the 4.0 C-rate for 3 minutes.

l. Examine the battery for the requirements of 3.10.5, 3.10.6, and 3.9.11.

4.5.30 Salt fog test. This test consists of the following steps:

a. Constant potential charge the battery in accordance with 4.4.2.

b. Open-circuit the battery.

c. Dissolve 0.91 kilogram (2 pounds) (nominal) of salt into 17.24 kilograms (38 pounds) (nominal) of distilled or demineralized water. The resistance of the water shall be not less than 500,000 ohms per centimeter. The salt shall contain not greater than 0.1 percent sodium iodide per weight and shall contain not greater than 0.5 percent total impurities. The pH of the solution shall range from 6.5 to 7.2. Add the solution to the salt fog test chamber and prepare the chamber for use. Stabilize the temperature of the chamber at 35 °C ±5 °C (95 °F ±3 °F).

d. Place not less than two salt fog collection receptacles in the chamber, with one placed nearest to the nozzle and one farthest from the nozzle. These shall be approximately 10 cm (4 in.) diameter and shall collect from 0.5 to 3 milliliters of solution per hour for every 80 square centimeters of horizontal collecting area in an average test of not less than 16 hours.

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e. Place the battery in the salt fog chamber. Elevate the battery above the floor of the chamber such that the bottom of the battery does not rest in any liquid.

f. Continuously atomize the salt solution into the test chamber for 48 ±2 hours. Measure the salt fog fallout rate and pH of the fallout solution at not less than 24-hour intervals.

g. Remove the battery from the salt fog chamber.

h. Open-circuit the battery for 8.5 ±7.5 hours.

i. Examine the battery for the requirements of 3.8.1.

j. Discharge the battery at the 4.0 C-rate for 3 minutes.

k. Examine the battery for the requirements of 3.8.1, 3.10.5, 3.10.6, and 3.9.11.

4.5.31 Temperature shock test. This test consists of the following steps:

a. Constant potential charge the battery in accordance with 4.4.2. Upon completion of charge, the battery shall remain on open-circuit through the completion of step 4.5.31.i.

b. Place the battery in a temperature chamber at 71 °C (160 °F) and expose the battery for 4.00 ±0.25 hours. At the end of this exposure period, lift the battery by its vent tubes.

c. At the conclusion of this time period, transfer the battery, within 5 minutes, to a chamber with an internal chamber temperature of -57 °C (-70 °F).

d. Expose the battery to -57 °C (-70 °F) for 4.00 ±0.25 hours. At the end of this exposure period, lift the battery by its vent tubes.

e. At the conclusion of this time period, transfer the battery, within 5 minutes, to a chamber with an internal chamber temperature of 71 °C (160 °F).

f. Expose the battery to 71 °C (160 °F) for 4.00 ±0.25 hours. At the end of this exposure period, lift the battery by its vent tubes.

g. Repeat steps 4.5.31.c through 4.5.31.f.

h. Repeat steps 4.5.31.c and 4.5.31.d.

i. Return the battery to room temperature and stabilize.

j. Discharge the battery at the 4.0C-rate for 3 minutes.

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k. Examine the battery for the requirements of 3.8.1, 3.10.5, 3.10.6, and 3.9.11.

4.5.32 Fluid susceptibility. This test consists of the following steps:

a. Conduct the fluid susceptibility test in accordance with MIL-STD-810, method 504 using fluids:

(1) Fuels: Kerosene aviation turbine fuel.

(2) Hydraulic oils: Mineral oil based.

(3) Lubricating oils: Ester based (synthetic).

(4) Solvents and cleaning fluids: Cleaning compound for aircraft surfaces.

b. Spray the battery with the appropriate fluid one or more times per day to maintain a wetted condition for a time not less than 24 hours.

c. Place the battery in a temperature chamber at 65 °C (149 °F) for 24 ±0.1 hours.

d. Return the battery to room temperature (see 4.4.1) and stabilize for 2 ±0.1 hours.

e. Examine the battery for the requirements of 3.8.1, 3.9.2, and 3.9.3.

f. Perform the tests of 4.5.4 and 4.5.8.

g. Examine the battery for the requirements of 3.8.1, 3.9.3, 3.10.5, 3.10.6, and 3.9.6.

4.5.33 Sand and dust test. This test consists of the following steps:

a. Conduct the sand and dust test in accordance with MIL-STD-810, method 510.4, procedure I.

b. Perform the tests of 4.5.2 and measure the dimensions of 4.5.3.

c. Examine the battery for the requirements of 3.8.l, 3.9.2, and 3.9.3.

d. Perform the tests of 4.5.4 and 4.5.8.

e. Examine the battery for the requirements of 3.8.1, 3.10.5, 3.10.6, 3.9.6, and 3.9.7.

4.5.34 Ground storage test. This test consists of the following steps:

a. Constant potential charge the battery in accordance with 4.4.2.

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b. Place the battery in a temperature chamber at 50 °C (122 °F) for 30 days on open-circuit.

c. After the storage period, remove the battery from the chamber and let it stand open-circuit for 8 ±2 hours at room temperature (see 4.4.1).

d. Discharge the battery at the 1.0 C-rate to 20.0 volts.

e. Examine the battery for the requirements of 3.9.16.

f. Repeat steps 4.5.34.a through 4.5.34.e but for step 4.5.34.b use a temperature of -40 °C (-40 °F).

g. Examine the battery for the requirements of 3.9.16.

4.5.35 Shelf life test. Constant potential charge the battery in accordance with 4.4.2. Store the battery for not less than 18 months in an uncontrolled temperature environment (such as an unheated, uncooled warehouse) where the ambient temperature may reach hot temperatures as specified for a Hot Dry condition, and cold temperatures as specified for Severe Cold condition, in MIL-STD-810. Periodic charging in accordance with 4.4.2 is permitted during storage at intervals of not less than 12-months. If the battery is stored below 0 ºC or above 40 ºC, then period charging in accordance with 4.4.2 is permitted during storage at intervals of not less than 6-months. At the end of the 18-month storage period, constant potential charge the battery in accordance with 4.4.2. Stabilize the battery at room temperature (see 4.4.1). Discharge the battery at the C-rate to 20.0 volts. Constant potential charge the battery in accordance with 4.4.2. Subject the battery to the cycling test of 4.5.11. Examine the battery for the requirements of 3.9.10 and 3.9.17.

4.5.36 Penetration test. Test in accordance with either step (a) or step (b), see applicable specification sheet.

a. Nail penetration shall be tested in accordance with 4.5.36.a(1) through 4.5.36.a(4).

(1) Insert a 2.5 mm diameter stainless steel nail in the battery under test. The following parameters shall be specified by the procuring activity: length of nail, location of nail penetration, and number of cells to be penetrated.

(2) Leave the nail in the battery for 24 hours.

(3) Monitor the voltage and temperature of the battery for the duration of the test.

(4) Examine the battery for the requirements of 3.9.14.

b. Bullet penetration shall be tested in accordance with 4.5.36.b(1) through 4.5.36.b(4).

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(1) Secure the battery to a mounting apparatus in the same way the battery will be mounted in the end application.

(2) Impact direction. The impact direction of the bullet shall be specified by the procuring activity.

(3) Fire three bullets of caliber specified by applicable specification sheet through the battery in the directions specified above. One of the bullets shall be a tracer bullet.

(4) Examine the battery in accordance with 3.9.14.

4.5.37 Crush. A battery is to be crushed between two flat surfaces. The force for the crushing is to be applied by a vise or by a hydraulic ram with a 32mm (1.25 in.) diameter piston. The crushing is to be continued until a pressure reading of 17.2 MPa (2500 psig) is reached on the hydraulic ram, applied force approximately 13 kN (3000 pounds). Once the maximum pressure has been obtained it is to be released. The orientation of the battery during the crush test shall be specified in the applicable specification sheet. Each sample battery is to be subjected to a crushing force in only one direction. Separate samples are to be used for each test Examine the battery in accordance with 3.9.15.

4.5.38 BMS tests.

4.5.38.1 BMS requirements test. Verify that the fully assembled breadboard un-potted battery has BMS electronic, cell strings, sensors, and heater blankets required as provided by manufacturers design description. Verify that that design meets requirements of 3.9.3, 3.10.1, and battery specification sheet (see 3.1) with BMS electronic, cell strings, sensors, and heater blankets included in the battery.

4.5.38.2 BMS power test. Verify that the BMS receives power during the following tests as applicable to 3.10.2.a and 3.10.2.b:

a. Powered from internal battery cells test. This test determines that when external power is not connected or during a deep discharge event (see 3.10.4), that the BMS is powered from internal battery cells.

(1) Constant potential charge the battery in accordance with 4.4.2. At the completion of the charge and while on open circuit, verify the BMS is receiving power from internal battery cells. Record the discharge rate (in amperes) that the BMS receives during this state. Then, disconnect the battery and verify that the BMS continues to receive power from the internal battery cells.

(2) Discharge the battery at 0.5 C-rate. When the battery voltage is discharged to 20.0 volts, verify that the BMS is powered by internal battery cells. Continue to

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discharge the battery at 0.5 C-rate for 30-seconds, and verify that the BMS continues to receive power from the internal battery cells throughout the 30-second duration.

(3) Continue to apply a constant-current load of 0.5 C-rate to the battery for an additional 30 seconds. In accordance with 3.10.4, the battery discharge should now be inhibited. Verify that the BMS continues to receive power from the internal battery cells.

b. Powered in parallel by the internal cells and external power source test. This test verifies that the BMS receives power in parallel by the internal cells and external power source when the battery is connected to an external power source.

(1) During the test of 4.5.38.3 and 4.5.38.15, verify that the BMS is receiving power in parallel by the internal battery cells and external power source.

(2) During the maintenance equipment compatibility test of 4.5.38.15, verify that the battery BMS is being powered by the battery charger/analyzer.

4.5.38.3 Electronics tests. The following electronics test shall be conducted on the battery and BMS:

a. Aircraft electrical power characteristics test. The guidelines for aircraft electrical power tests for 28 VDC bus can be found in MIL-HDBK-704-1 and for other aircraft bus types in accordance with applicable part of MIL-HDBK-704-1. This test consists of the following steps:

(1) Constant potential charge the battery in accordance with 4.4.2.

(2) At ambient temperature discharge the battery at the 1.0C-rate to a cutoff voltage of 20.0 volts.

(3) Constant potential charge the battery in accordance with 4.4.2.

(4) The test setup may include the battery, a battery charger (power source – converter, etc.), and DC load all connected to a common bus. Testing can be conducted for two modes of operation, discharging at a 5-ampere rate with the charger off and charging/floating on the load bus at any rate of charge with the charger on. The battery should remain on and connected to the load bus throughout the test and should not disconnect from the bus.

(5) Functionality of the battery should be demonstrated and recorded before and after each aircraft electrical system operating condition by operating by first discharging the battery at the 1.0 C-rate for 60.0 + 0.0 / - 0.1 minutes or to cutoff

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voltage of 20.0 volts, followed by a constant potential charge in accordance with 4.4.2.

(6) Following completion of the testing steps 4.5.38.1 through 4.5.38.3, the test should be repeated.

(7) Examine battery for the requirements of 3.10.3 3.10.4, and 3.10.5.

b. Battery BMS protective circuits test. Visually inspect the battery to verify that the battery BMS protective circuits, including battery disconnect function, are suitably protected from cell failure conditions within the battery system such that the safety function is not compromised. Visually inspect the battery to verify that battery BMS electronics are located in a separate compartment from the cell stacks.

c. Battery BMS warning circuits test. Visually inspect the battery to verify that the battery BMS warning circuits, including fault and built-in test (BIT) functions, are suitably protected from cell failure conditions within battery system such that the warning signal is not compromised. Visually inspect the battery to verify that battery BMS electronics are located in a separate compartment from the cell stacks.

4.5.38.4 Deep discharge recovery tests. This test consists of the following steps:

(a) Constant potential charge the battery in accordance with 4.4.2.

(b) Discharge the battery at 0.5 C-rate at room temperature (see 4.4.1).

(c) Verify that the protective circuitry of the battery inhibits discharge once the battery pack voltage is discharged to 20.0 volts or less and following the time delay (see 3.10.4).

(d) Place the battery on open circuit for a period of 5.0 ± 0.5 minutes. Verify that the battery remains offline (no on-off-on-off chatter) during this time period.

(e) Apply a constant-current load at 0.5 C-rate to the battery and verify that the protective circuitry continues to inhibit discharge of the battery.

(f) Constant potential charge the battery at 28.25 ± 0.25 volts until the battery pack voltage rises to 24.25 ± 0.25 volts.

(g) Discharge the battery at 0.5 C-rate and verify that the protective circuitry allows for the discharge of the battery in accordance with 3.10.4.

(h) Constant potential charge the battery in accordance with 4.4.2.

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(i) Repeat step (b) using the engine start, emergency, or fault clearing load profile, whichever represents the most demanding loading scenario, from the applicable specification sheet.

(j). Repeat step (c) and step (h).

(k) Discharge the battery at 0.5 C-rate at room temperature to a voltage of 20.0 + 0.0 / -0.1 volts and immediately cutoff the load.

(l) Place the battery on open circuit for a period of 10.0 ± 1.0 seconds.

(m) Apply a constant-current load at 0.5 C-rate and verify that the protective circuitry allows for the discharge of the battery for at least 30 seconds before discharge is inhibited in accordance with 3.10.4.

4.5.38.5 State of charge, state of health, and fault indicator tests.

a. SOC indication test. This test consists of the following steps:

(1) Constant potential charge the battery in accordance with 4.4.2.

(2) Discharge the battery at 1.0 C-rate corresponding to the BOL capacity to 20.0 volts. Measure the discharge capacity and record the voltage profile of the battery.

(3) Constant potential charge the battery in accordance with 4.4.2.

(4) Discharge the battery at 1.0 C-rate corresponding to the BOL capacity to 20.0 volts. Measure the discharge capacity and record the voltage profile of the battery.

(5) Verify the discharge capacity values derived in step (2) and step (4) are within 1percent of each other.

(6) Use the lowest value of discharge capacity between step (2) and step (4) as the Measured Full Charge (MFC) capacity of the battery.

(7) Constant potential charge the battery in accordance with 4.4.2. After this charge, the capacity of the battery is estimated to be 100 percent of the MFC, denoted as MFC_1.

(8) Based on the BMS design documentation provided by the manufacturer, correlate the MFC_1 capacity to the expected SOC value to be shown on the digital display. This expected SOC value may be denoted as SOC_1.

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(9) Verify that the SOC_1 value matches the SOC value shown on the digital display with an accuracy of ± 5 percent.

(10) Discharge the battery at a rate of 1.0 C-rate with respect to the MFC capacity for 12 minutes. After this discharge, the capacity of the battery is estimated to be 80% of the MFC, denoted as $MFC_{0.8}$.

(11) Based on the BMS design documentation provided by the manufacturer, correlate the $MFC_{0.8}$ capacity to the expected SOC value to be shown on the digital display. This expected SOC value may be denoted as $SOC_{0.8}$.

(12) Verify that $SOC_{0.8}$ value matches the SOC value shown on the digital display with an accuracy of ± 5 percent.

(13) Repeat steps (10) through (12) in intervals of 12 minutes such that values for 60% MFC ($MFC_{0.6}$), 40% MFC ($MFC_{0.4}$), 20% MFC ($MFC_{0.2}$), and 0% MFC (MFC_0) are derived and associated with $SOC_{0.6}$, $SOC_{0.4}$, $SOC_{0.2}$, and SOC_0.

b. SOH indication test. This test consists of the following steps:

(1) Use the MFC capacity value derived in 4.5.38.5.a.(6) to calculate the battery SOH in accordance with 3.10.5.b.

(2) Verify that the SOH value shown on the digital display is within ± 5 percent of the calculated SOH from 4.5.38.5.b.(1).

c. Fault indication test. This test consists of the following steps:

(1) Using a breadboard battery (see 6.15.16), induce a fault condition corresponding to an under-voltage fault.

(2) Verify that the fault condition is detected by the BMS software and a fault message is shown on the digital display in accordance with table IX.

(3) Verify that the BMS software responds to the fault condition in accordance with table IX.

(4) Using the J2 connector, verify that the fault message has been stored in volatile memory and is linked to an accurate time stamp.

(5) Disconnect the battery from any load or charge voltage and wait one minute.

(6) Activate the battery by providing a charge voltage in accordance with 4.4.2 or as specified in the applicable specification sheet.

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(7) Using the J2 connector, verify that the fault message has been stored in non-volatile memory and is linked to an accurate time stamp.

(8) Verify that the fault message is shown on the digital display in accordance with table IX.

(9) Repeat steps (1) through (8) for each other fault condition shown in table IX.

4.5.38.6 BIT test. This test consists of the following steps performed on the breadboard battery:

a. Constant potential charge the battery in accordance with 4.4.2. During the charging, perform the following steps:

b. Verify that the BMS executes BITs upon the start of the usage even in the previous step.

c. Simulate a fault condition on the breadboard battery and verify that the fault has been identified by the associated BIT and reported to the digital display when the PB is activated.

d. Perform 20 trials of each fault condition shown in table IX and verify that the BIT diagnostic probability for each fault condition is 95 percent or greater.

4.5.38.7 Voltage monitoring test. This test consists of the following steps:

a. Using a breadboard battery (see 6.15.16), perform the test of 4.5.8 and verify that the cell, string, pack, and terminal voltages are being monitored and stored in accordance with 3.10.14, table IV, and table V.

b. Verify that for cells in a parallel electrical connection, if applicable, are at the same potential, which is monitored and stored as a single group.

c. Verify that the sample rate of cell, string, pack, and terminal voltages are in accordance with 3.10.6.

d. Verify that the BMS references the upper and lower voltage limits stored in non-volatile memory for detecting faults associated with cell, string, pack, and terminal voltages.

e. Using a breadboard battery (see 6.15.16), perform the test of 4.5.8 and simulate a battery cell that has violated the associated upper voltage limit (see 3.10.7.d). Verify that the BMS detects and responds in accordance with 3.10.7.e and table IX. Repeat this step with a violation of the associated lower voltage limit (see 3.10.7.d) and verify that the BMS generates a fault and responds in accordance with 3.10.4 and table IX.

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(1) Repeat step (b) with respect to battery string voltage.

(2) Repeat step (b) with respect to battery pack voltage.

(3) Repeat step (b) with respect to battery terminal voltage

4.5.38.8 Temperature monitoring tests. This test consists of the following steps:

a. Using the breadboard battery verify the amount and location of cell and battery sensors required for safe operation of the battery during the battery's service life.

b. During the tests conducted in 4.5.38.7.a, verify that the cell and battery temperature sensor readings are being monitored and stored in accordance with 3.10.8, table IV, and table V.

c. Determine the operating range for the cell chemistry used in the battery and temperatures that shall not be exceed during charge and discharge.

d. At the end of charge and discharge test conducted in 4.5.38.7.a, heat one cell past its temperature limit. Verify that the BMS uses the temperature limits programmed into the non-volatile memory to detect an over-temperature fault. Verify that the BMS detects and responds to the fault in accordance with 3.10.7.e and table IX.

e. Verify that the cell string in 4.5.38.8.d, where the cell temperature was exceeded, has been inhibited from charge or discharge by the BMS in accordance with table IX.

f. Verify that the temperature fault is displayed when the push button is pushed in accordance with table IX. Verify that the temperature fault data is being stored properly in accordance with 3.10.7.e.

4.5.38.9 Current monitoring test. Prior to conducting this test replace the cell string in the breadboard battery that are unsafe to charge or discharge. This test consists of the following steps:

a. During the tests conducted in 4.5.38.7.a, verify that the battery string and pack current are being monitored and stored in accordance with 3.10.14, table IV, and table V.

b. Verify that the maximum charge and discharge current reported by the battery manufacturer is capable of satisfying the emergency, engine start, and fault clearing load requirements shown in the applicable specification sheet. Verify that these limits are stored in non-volatile memory.

c. Using the breadboard battery, perform the following test:

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(1) During steps (h) through (i) of 4.5.38.4, verify that the BMS does not inhibit discharge or charge of the battery in accordance with 3.10.4.

(2) Constant potential charge the battery in accordance with 4.4.2.

(3) Discharge the battery at a discharge current exceeding the limit required in 3.10.9.b and verify that the BMS detects and responds to the over-current on discharge condition in accordance with table IX.

(4) Place the battery on open circuit for a period of 5.0 ± 0.5 minutes.

(5) Discharge the battery at the 1.0 C-rate to approximately 50 percent SOC.

(6) Charge the battery at a charge current exceeding the limit required in 3.10.9.b and verify that the BMS detects and responds to the over-current on charge condition in accordance with table IX.

d. Using the breadboard battery (see 6.15.16), simulate a battery string with an over-current on charge condition and verify that the BMS inhibits or limits the charge current of that string while the remaining battery strings are allowed to continue charge and discharge. Repeat this verification for the over-current on discharge condition.

4.5.38.10 Balance monitoring test. Prior to conducting this test replace the cell string in the breadboard battery that is unsafe to charge or discharge. The BMS balancing capability shall be as follows:

a. Verify that the BMS performs balancing of cells, strings, or modules as appropriate to maintain battery capacity, life, and safety by performing the following test with the breadboard battery:

(1) Constant potential charge the battery in accordance with 4.4.2.

(2) Discharge the battery at a 1.0 C-rate to 50 percent ± 5.0 percent SOC.

(3) Place the battery on open circuit for a period of 5.0 ± 0.5 minutes.

(4) Induce a 55.0 ± 5.0 mV imbalance in a cell, string, or module.

(5) If the battery is top-balanced, then constant potential charge the battery in accordance with 4.4.2 and verify that all cells are balanced. If the battery is bottom-balanced, then discharge the battery at a 1.0 C-rate to approximately 0% SOC or lower voltage cutoff and verify that all cells are balanced.

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b. Refer to the applicable specification sheet for the balancing load profile, and verify that the BMS is capable of balancing the cells within the first hour of flight. If the balancing load profile is not available, then verification is satisfied through 4.5.38.10.a.

c. During the test of 4.5.38.10.a, verify that the BMS monitors and stores cell balancing, string balancing or module balancing, and battery capacity in accordance with 3.10.14, table IV, and table V.

d. Verify by reviewing the end of charge data stored by the BMS that the difference between the maximum and minimum cell voltages does not exceed 50mV during the charge test of 4.5.38.10.a.

e. Using the discharged breadboard battery jumper around a cell, string, or module and conduct a charge test in accordance with 4.5.7. Fifty-five minutes into the charge, remove the jumper and verify that the BMS was unable to balance the cell, string of cells, or a module. Verify that the BMS detects and responds to the fault in accordance with 3.10.7.e and table IX.

f. Using the breadboard battery (see 6.15.16), simulate a balancing over-temperature condition and verify that the BMS detects and responds to this fault in accordance with 3.10.7.e and table IX.

4.5.38.11 Short circuit and in-rush current protection test. Battery and BMS short circuit test shall be as follows:

a. Continuously monitor and record voltage, current, pressure, and temperature. Bypass all external and accessible internal electrical safety devices and short the test sample through a load of not greater than 0.02 ohm. The load shall remain attached for not less than 24 hours. Verify that the test sample met the requirements of 3.8.1.a and 3.8.1.c.

b. Verify that the components in the charge and discharge path withstood short-circuit currents by conducting a charge, discharge, and charge in accordance with 4.5.7.

c. Verify BMS has limited in-current to cells and electronic by checking for current and BIT faults on the display. Conduct emergency discharge flight profile test in accordance with the battery specification sheet (see 3.1).

d. During the test of 4.5.38.11.a, verify that the BMS detects and responds to the external short-circuit condition in accordance with 3.10.7.e and table IX.

e. Using the breadboard battery (see 6.15.16), simulate an internal short-circuit condition and verify that the BMS detects and responds to this fault in accordance with 3.10.7.e and table IX.

4.5.38.12 Data storage. This test consists of the following sections:

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a. Frequency. Using the breadboard battery, perform the following tests.

(1) Initiate a usage event (see 6.15.17) of the battery and verify that the battery parameters specified in table IV have been stored in memory within 1 second of the start of the usage event.

(2) Verify that the battery parameters specified in table IV continue to be stored in memory at a rate of every one minute or faster.

(3) Induce one of the fault conditions specified in table IX and verify that the parameters specified in table IV are stored in memory at a frequency of every 10-seconds or faster.

(4) End the usage event.

(5) Constant potential charge the battery in accordance with 4.4.2 and verify that the battery parameters specified in table IV are being stored in memory at a frequency of every one minute or faster.

(6) Discharge the battery at 0.5 C-rate at room temperature to induce a deep discharge condition. During the 30-second time delay, verify that the data storage frequency is at every 10 seconds or faster.

(7) Repeat step (5).

b. Storage format.

(1) Constant potential charge the battery in accordance with 4.4.2.

(2) Discharge the battery at 1 C-rate for 10 minutes.

(3) Disconnect the load from the battery and rest at open circuit voltage for 10 minutes.

(4) Retrieve the data UEF containing the parameters specified in table IV using a non-proprietary PC-based tool and verify that the data is formatted in accordance with table V.

(5) Verify that the UEF contains data associated with the last five (5) usage events.

c. Software and Fault Limits. Verify the software and fault limits are stored in non-volatile memory.

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d. Historical performance data. Following the tests of 4.5.38.12.b, verify that the BMS has stored, in a separate file distinct from other data files, the historical performance data in accordance with table V.

e. Historical fault data. Following the tests of 4.5.38.12.b, verify that the BMS has stored, in a separate file distinct from other data files, the historical fault data in accordance with 3.10.7.e.

f. Storage capacity.

(1) Constant potential charge the battery in accordance with 4.4.2.

(2) Discharge the battery at the 0.5 C-rate for 30.0 ±1.0 minute.

(3) Repeat steps (1) and (2) until 420.0 minutes +1.0 minute/- 0.0 minute have elapsed and verify that the BMS has recorded 7 hours of data at a frequency in accordance with 3.10.12.a.

(4) If applicable, continue repeating steps (1) and (2) until 450.0 minutes +1.0 minute/-0.0 minute have elapsed, and verify that the BMS has maintained at least 7.0 hours of data for the usage event composed of the latest data points.

(5) Measure the file size for 7 hours of data and verify that BMS memory is sufficient to store at least a total of five files of the same file size in non-volatile memory.

4.5.38.13 External communications test. External communications test shall be as follows:

a. Using an external portable computer, connect the battery and verify that the data stored in memory is transferrable and in the format defined in table V.

b. Verify that communication from the battery was accomplished through the sensor receptacle J2.

4.5.38.14 BMS resources test. This test consists of the following steps:

a. For each test that utilizes a breadboard battery in section 4, record the clock speed of the BMS processor during that testing.

b. Verify the maximum clock speed measured in step (a) is 50 percent or less than the maximum clock speed possible for the BMS processor described in the BMS design document (see 3.10, 6.2.i) provided by the battery manufacturer.

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c. For each test in section 4, measure the file size of the combination of BIT logs, historical performance data, and historical fault data stored in memory.

d. Estimate the maximum combined file size from step (c) and project the growth of that file size over a 5-year period.

e. Verify that the maximum file size estimated in part (d) is 50 percent or less than the maximum storage capacity for the BMS memory described in the BMS design document (see 3.10, 6.2.i) provided by the battery manufacturer.

4.5.38.15 Maintenance equipment compatibility test. Maintenance equipment compatibility test shall be as follow:

a. Using Operating Instructions, for Battery Charger/Analyzer, P/N: MSD-970-1 or RFK-80 program the Battery Charger/Analyzer, P/N: MSD-970-1 or RFK-80 to charger, discharge, and analyze the battery using recommended charge and discharge rates and times to perform proper maintenance from 3.10.17.a.

b. Connect the battery to the MSD-970-1 charger/analyzer or RFK-80 and allow it to perform maintenance.

c. Verify there is no incompatibility with the MSD-970-1 charger/analyzer or RFK-80 and the battery.

d. Verify that the battery meets requirements for capacity per specification sheet (see 3.1).

4.5.39 Final examination. Without dissecting the battery, examine it for the requirements of 3.8.1. Subject the battery to the dielectric strength test of 4.5.5. Examine the battery for the requirements of 3.9.4.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The storage batteries covered by this specification are used in fixed-wing and rotary-wing military aircraft. The batteries are used for starting engines or auxiliary power units. They also provide electricity to support aircraft emergency loads, aeronautical and ground systems, and associated equipment, have non-removable covers, and are designed for maintenance-free operation (see 6.15.8). Most of the batteries under this specification are designed to be charged directly from the aircraft DC electrical bus (see 6.10.5). However, some systems may have a separate charger that either is contained within the battery or is external to the battery. The batteries are also exposed for prolonged periods to extreme seagoing environments not encountered by civilian aircraft.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification and applicable specification sheet.

b. Packaging requirements (see 5.1).

c. Battery required, including quantity, NSN, and part or identifying number (PIN; see 6.16).

d. Qualification testing and conformance inspections are required as indicated herein (see 4.2, 4.3.3, 6.8, table VI and table VII for the quantity of samples required).

e. Whether certification of conformity is required (see 3.3).

f. Shelf-life coding for an 18-month shelf life period (see 3.9.17).

g. Responsibility for inspection and sample selection (see 6.8).

h. If certification to S9310-AQ-SAF-010 is required (see 3.8.8).

i. Manufacturer will provide a report, if required in the contract or purchase order, that provides a description of battery and BMS design that include the following information:

(1). Protective circuits operation and protection methods.

(2). Warning circuits operation and protection method.

(3). Description of how software estimates the battery SOC and SOH.

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(4). Fault detection and display method.

(5). BIT detection and display method.

(6). Description of the number of cells, configuration of cells, orientation of cells, cell chemistry, cell voltage, string voltage, pack voltage, terminal voltage, number of strings, open circuit voltage, voltages being monitored and stored, voltage data storage format.

(7). Description of voltage reference checks for cell voltage and battery voltage monitoring.

(8). Description of sampling rate adjustment capability and set points to assure the battery performance and safety.

(9). Upper and lower voltage limit for the cell, string, pack, and terminal elements.

(10). Number and placement of temperature sensor, cell chemistry operating temperature range, and cell chemistry temperature limits.

(11). Charge and discharge data storage format, and maximum charge and discharge current limits for the cell chemistry and design.

(12). Charge and overcharge data storage format and charge and overcharge limits for the cell chemistry and design.

(13). Discharge and over-discharge data storage format and discharge and over-discharge limits for the cell chemistry and design.

(14). Description of cell balancing method, accuracy, and fault indications.

(15). Description of data storage in memory and data storage format.

(16). BMS processor description, processing bandwidth, non-volatile program memory amount and usage, and volatile and non-volatile storage amount and usage.

(17). Manufacturer recommended charge and discharge rates and times for the battery chemistry and design to use in fleet maintenance shops equipped with the following battery charger/analyzers: MSD-970-1, RF80-K, and RF80-M.

6.3 Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313. Note that 29 CFR 1910.1200 requires that the Safety Data Sheet for each hazardous chemical

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used in an operation must be readily available to personnel using the material. Contracting officers will identify the activities requiring copies of the Safety Data Sheet.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 29595, whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Crane Division, Naval Surface Warfare Center, Power & Energy Division, Airborne and Space Energy Systems Branch, Code JXMO, 300 Highway 361, Crane, IN 47522-5001 or the Electrical Power Systems Division, AIR-4.4.5.2, Bldg. 1461, Naval Air Warfare Center Aircraft Division, 48298 Shaw Road, Unit 4, Patuxent River, MD 20670-1900. The preferred qualification testing facility is the Crane Division, Naval Surface Warfare Center, Power & Energy Division, Airborne and Space Energy Systems Branch, Code JXMO, 300 Highway 361, Crane, IN 47522-5001. An online listing of products qualified to this specification may be found in the qualified products database (QPD) at https://assist.dla.mil.

6.5 Application for qualification. Applicants obtain the approval of the qualifying activity before submitting samples for qualification inspection. The process starts when the manufacturer applies for qualification by letter to the qualifying activity indicated in 6.4. SD-6 “Provisions Governing Qualification” provides guidance concerning the necessary information and certification (see 6.10.7). Copies of the SD-6 are available from the ASSIST database at https://assist.dla.mil.

6.6 Conformance inspection. Conformance inspection by the qualifying activity will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of conformance inspection by the qualifying activity may be adjusted to make maximum use of the contractor's quality control system and the quality history of the product (see 4.3). Any or all qualification tests may be performed on group B samples if the manufacturer has not supplied the product to the procuring activity for a period greater than 2 years, or if the qualifying activity suspects that the product no longer conforms to prescribed requirements. When only one sample is received for group B inspection, the qualifying activity will decide which one of the test sequences indicated by the specification sheet will be performed. If a sample does not meet the requirements of 3.9.2, the qualifying activity may terminate further test and inspection. The preferred conformance testing facility is the Crane Division, Naval Surface Warfare Center, Energy Power & Energy Division, Airborne and Space Energy Systems Branch, Code JXMO, 300 Highway 361, Crane, IN 47522-5001.

6.7 Group B costs. An estimate of the qualifying activity's cost for testing for single service procurements, for use by a contractor at time of preparation of bids, may be obtained

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from the Crane Division, Naval Surface Warfare Center, Power & Energy Division, Airborne and Space Energy Systems Branch, Code JXMO, 300 Highway 361, Crane, IN 47522-5001.

6.8 Responsibility for inspection and sample selection. The qualifying activity (see 4.2, 6.2, and 6.4) performs qualification inspection on samples furnished by the contractor. The contractor performs the inspections specified for group A conformance inspection (see 4.3.4). The Government selects samples for group B conformance inspection in accordance with 4.3.3. The qualifying activity conducts group B conformance inspection (see 4.3.5). The Government reviews and examines the contractor's inspection procedures and inspection records as necessary (see 6.2.d and 6.2.g).

6.9 Charge/discharge rate discussion. The designated discharge rate is determined by multiplying the nominal rated capacity (C-rating) (see 6.15.1) in ampere-hours times the rate of discharge factor listed in a test. Some examples of charge and discharge rates (see 6.15.2) are explained below. A battery undergoing a 1C or C-rate discharge must produce a current equal to its ampere-hour capacity for at least 60 minutes at a specified temperature before its potential drops below the cutoff or end voltage. For example, a battery with a nominal capacity rating of 35.0 Ah/1 hr/25 °C/20.0 V undergoing a 1C or a C-rate discharge must produce a current of:

1C or C-rate	=	1 × 35 ampere-hours
	=	35 amperes for at least 60 minutes at 25 °C ambient temperature before the battery potential drops below the cutoff or end voltage of 20.0 volts

The charge or discharge may take place over a period other than 60 minutes or may instead continue until a desired voltage level is reached. For example, a 35-ampere-hour battery charged at C/5 for 3 hours would receive:

C/5-rate	=	35 ampere-hours ÷ 5
	=	7 amperes delivered to the battery for 3 hours

6.10 Design practices and lessons learned.

6.10.1 Cell or battery compartment. A special compartment must be designed for the cell or battery for isolation from the electronics of the end item equipment or weapon system. The compartment must have no interior projections nor sharp edges that could damage the electrical insulation around the cell or battery. The cell or battery must be secured within the compartment to reduce shock and vibration to the levels required for end item use. The compartment must be designed either to provide vent paths or else contain gases emitted by a venting battery. If vent paths are to be used, they must be identified on the compartment drawings. If the compartment is designed to contain the vented gas, the drawings must identify the pressure-containing capabilities of the compartment (including pressure level) and the containment designs or methods.

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6.10.2 Switches. Battery or cell switches in the end item equipment or weapon system must be carefully selected to prevent accidental cell or battery turn-on. No switching may occur in the ground leg(s).

6.10.3 Cell or battery dissimilarity. Cells or batteries with different physical characteristics, chemistries, or electrical parameters must not be used concurrently in the same electrical circuit.

6.10.4 Dissimilar metals. MIL-STD-889 provides methods for protecting joined dissimilar metals (see 3.4).

6.10.5 Charging the battery from the aircraft DC bus on the aircraft. Due to such factors such as internal impedance, polarization effects, and gassing during charging, batteries do not charge at 100 percent efficiency. Therefore, a nominal 26-volt battery needs a potential of 28.0 ±0.5 volts for proper charging. The period of time required to reach full charge is affected by charging potential and battery temperature. Charging while at the lower end of the potential range will require a longer period of time than at the higher end. As battery temperatures decrease, higher charging potentials are required for optimum charging efficiency. The characteristics of power provided by the aircraft DC bus, as well as aircraft power characteristics in general, are often specified in accordance with MIL-STD-704. When determining whether the battery can be adequately charged from the bus, designers must consider how heavily the DC bus is loaded, what percentage of the time the potential is outside the charging range, how often batteries will be charging versus supplying the bus, the potential for load growth on a bus that is currently satisfactory, the ambient temperature of the battery compartment, and similar situations. If the power provided to the battery is not sufficient for proper charging, special arrangements (such as a charger or dedicated transformer/rectifier) will be required to ensure that batteries will be properly charged within an acceptable period of time.

6.10.6 Additional battery information. See NAVSO P-3676, Navy Primary and Secondary Batteries: Design and Manufacturing Guidelines, for additional information on batteries. Copies are available from Customer Service (Code 033343), NAVSUP Weapon Systems Support, 700 Robbins Avenue, Philadelphia, PA 19111.

6.10.7 Additional qualification information. See SD-6 for guidance on obtaining qualification. The document describes the manufacturer's obligations to:

a. Maintain process and quality control procedures that ensure that the items comply with all specification requirements.

b. Immediately report any discrepancies disclosed during testing or periodic reexamination of the product and his production process controls.

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c. Ensure that delivered items conform to all specified product characteristics (in no way relieving manufacturers of contractual obligations to ensure delivered items comply with all specification requirements).

6.11 Test equipment and facilities. Measurements should be performed (see 4.5) with equipment that has the following characteristics. Test equipment and inspection facilities should have sufficient accuracy, quality, and quantity to permit performance of the required inspection. Test and measurement equipment should be calibrated and traceable to the National Institute of Standards and Technology. All voltmeters and ammeters should be accurate within ±0.5 percent of the full scale reading. The range of analog type meters should be such that the readings are taken on the upper half of the scale. Timers should be accurate within ±0.5 percent. The sensitivity of voltmeters should be not less than 5000 ohms per volt. In all tests involving discharge through a resistance, the total circuit resistance should be accurate within ±1 percent. For all discharges, the current or voltage (whichever is applicable) should be maintained within ±1 percent of the specified value at all times, and the duration of the discharges should be within ±2 percent of the nominal discharge time. All discharges will be continuous unless otherwise stated.

6.12 Safety and storage. Lithium batteries with aluminum battery containers should not be maintained in the same area as vented nickel-cadmium batteries. The potassium hydroxide electrolyte of the nickel-cadmium battery will damage aluminum. However, lithium batteries with aluminum battery containers may be stored together. S9310-AQ-SAF-010 contains additional storage procedures

6.13 Disposal. End users will comply with federal, state, and local regulations and S9310-AQ-SAF-010 for disposal of used batteries. Such items should be sent to the end user's local Defense Reutilization and Marketing Office, which will dispose of the items via contracts for recycling or other means.

6.14 BMS electronics. BMS electronic placement and application within the battery should take into account that a venting of a cell should not compromise the safety of the battery during flight. Things that have been found to work are placing them in separate compartments from the cell stacks. Sensors circuit cards within the cell stacks should have coatings that are resistant to the cells electrolyte.

6.15 Definitions.

6.15.1 Capacity. A measure of the energy content of a battery, usually stated in terms of ampere-hours. Capacity is computed by multiplying the number of amperes flowing from a cell or battery on discharge (at a given temperature and down to a specified end voltage) by the time in hours the cell or battery will deliver the amperes of current (see 3.9.7 and 6.15.9). An example of a shorthand way of stating capacity is 35.0 Ah/1 hr/25 °C/20.0 V, which means the battery provides not less than 35.0 ampere-hours of capacity when continuously discharged at

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35 amperes for not less than 1 hour while maintaining a voltage of not less than 18.0 volts at the battery terminals, all of which occurs in an ambient air temperature of 25 °C.

6.15.2 Charge or discharge rate. The amount of current and length of time required to charge or discharge a battery or cell (see 3.9.7 and 6.9). Unless otherwise specified herein or in the specification sheet, the rate is expressed as xC, where x is a multiplier and C is the capacity required by the specification sheet. When stated as the C-rate, x equals 1.

6.15.3 Cut-off voltage. A specified voltage during the discharge of a cell or battery at which point the discharge will be terminated (see 3.9.7 and 4.5.8).

6.15.4 Cycle. A combination of a charge and discharge (see 3.9.10).

6.15.5 Fully discharged battery. A fully discharged battery or cell will have been discharged to the cutoff voltage. Such batteries and cells are capable of recovering the full capacity specified by the applicable specification sheet (see 3.10.4).

6.15.6 Initial voltage delay. Initial voltage delay is the time required at the beginning of discharge for the cell or battery to reach minimum voltage after the load is applied (see 3.9.7).

6.15.7 Inspection lot. The quantity of batteries, harnesses, or components of any one type, produced at any one place of manufacture, submitted at one time to conformance inspection (see 3.7.6 and 4.3.2).

6.15.8 Maintenance-free. Lithium batteries furnished under this specification should not require any maintenance beyond charging to restore full state of charge to the battery after being placed for extended periods of time in storage. See 3.9.17. The battery supplier should document the preferred storage conditions and the acceptable duration to recover > 90% of the rated BOL capacity after full recharge. The battery supplier should also suggest maintenance inspections and frequency along with top-off charge internals with the requirement voltage and current supply conditions; identify SOC and SOH features and their operational use, or means to facilitate a discharge capacity test.

6.15.9 Nominal rated capacity (C-rating). The capacity listed on the battery identification marking in accordance with the specification sheet; sometimes called end-of-life or minimum capacity (see 6.9). Note that when batteries and cells produce less than this amount of capacity (see 3.9.7 and 6.15.1) after testing and treatment in accordance with the appropriate military technical manual, the batteries and cells are considered to be at the end of their useful life and must be properly disposed.

6.15.10 Normal upright position. The battery cover is on top of the battery with respect to the earth's surface (see 4.5.14, 4.5.26, 4.5.27, and 4.5.28).

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6.15.11 State of charge (estimated). The estimated state of charge is a calculated estimate reported by the battery. The estimated state of charge is the ratio of (i) the measured capacity to (ii) the beginning of life (BOL) capacity of the battery multiplied by the battery SOH (see 6.15.12). Measured Capacity is the estimated amount of capacity remaining in a battery at some time as measured by the BMS. The Beginning of Life (EOL) capacity is the full charge capacity of the battery at its beginning of life state.

6.15.12 State of health (estimated-percent). The estimated SOH is defined as the ratio of the present full charge capacity of the battery at normal conditions to the beginning of life (BOL) capacity of the battery.

6.15.13 Unit. One single, individual battery (see 4.3.2).

6.15.14 Digital Display Push Button (PB). The digital display push button is only used to activate the digital display. The digital display should automatically deactivate after a period of time, as defined by the end user. The PB is not used to control the on/off state of the battery.

6.15.15 Beginning of Life (BOL) Capacity. The beginning of life capacity is the full charge capacity of the battery at the time that battery pack assembly has been completed by the battery manufacturer. The BOL capacity represents the capacity of the battery at the 100% SOH condition.

6.15.16 Breadboard battery. The breadboard battery is an un-potted version of the battery designed to this specification and submitted for qualification testing. The breadboard battery includes BMS electronics, software, cell strings with sensors, and heater blanket(s). The breadboard battery also includes any sensors or wiring required to simulate a test condition or measure a parameter as applicable to the specific requirement and verification.

6.15.17 Usage Event. A usage event of the battery is defined as starting when either load or a charge voltage is applied to the battery and ends when a load or charge voltage is removed from the battery and following a 1-minute time delay. If the load or charge voltage is present at the battery terminals, and if the BMS is inhibiting discharge or charge, then the usage event continues to be active until that load or charge voltage is removed from the battery terminals.

6.16 Part or identifying number (PIN). The PIN (see 6.2.c) for batteries acquired to this specification is created as follows: M29595/X-Y with "M29595" representing the specification number, "X" signifying the specification sheet number, and "Y" indicating any variation of the battery on a particular specification sheet.

6.17 Specification sheets. Each specification sheet should include as a minimum the following information:

a. Part or identifying number (PIN) (6.16).

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b. Shape and size of cell or battery (see 3.9.3).

c. Weight (see 3.9.3).

d. Terminal labeling (see 3.7).

e. Capacity requirements (see 3.9.7).

f. Sequence of tests (see 4.5).

6.18 Safety.

6.18.1 Lithium cell and battery safety. NAVSEA technical manual S9310-AQ-SAF-010 gives requirements for lithium cell or battery safety testing. This manual also requires the development of a safety data package prior to approval of the testing by the Naval Ordnance Safety and Security Activity. NAVSEA instruction 9310.1 establishes the policy and responsibilities for the safety of all lithium batteries and equipment powered by these batteries used in Naval applications. Adherence to the policies contained in NAVSEA instruction 9310.1 is required when testing batteries in accordance with S9310-AQ-SAF-010. A description of the lithium cell- or battery-powered system is also required prior to testing approval. NAVSEA letter 8020 Serial Number N84/521 dated 2 Apr 09 provides interim guidance and changes to NAVSEA instruction 9310.1. NAVSEA SG270-BU-SAF-10 applies for all batteries above 1killowatt. Copies of NAVSEA instruction 9310.1 are available on the Internet at http://www.navsea.navy.mil/Resources/Instructions/. Copies of the NAVSEA letter are available from the Naval Ordnance Safety and Security Activity (Code N841), 3817 Strauss Avenue, Suite 108, Indian Head, MD 20640 or http://www.navsea.navy.mil/Resources/Instructions/.

6.18.2 System testing. Lithium cell or battery safety testing is specifically designed to force the lithium cells or batteries into an abused condition which will most likely cause the cells or batteries to swell, rupture (vent), catch on fire, or explode. After analyzing the safety data package, the Naval Ordnance Safety and Security Activity will determine whether the lithium cells or batteries must be installed into the actual lithium cell- or battery-powered device or weapon system while undergoing the abusive testing. Therefore, program managers must be aware of the necessary hardware requirements and plan accordingly.

6.18.3 FMECA testing. Failure mode effects and criticality analysis (FMECA) is used to systematically evaluate and document, by item failure mode analysis, the potential impact of each functional or hardware failure on mission success, personnel and system safety, system performance, maintainability, and maintenance requirements. The SAE standard ARP5580 can be used for guidance on FMECA testing.

6.18.4 Software certification testing. Software certification is used in critical systems that must meet minimum safety, security, and reliability standards. The battery software will need to interact with the aircraft software in order to provide state-of-health information, for this

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reason software certification testing may be necessary. The RTCA DO-178 can be used for guidance on software certification testing. The qualifying activity will issue a letter of authorization for qualification stating the requested software.

6.19 Transportation. Official Department of Transportation guidance is provided in 49 CFR. Official Navy guidance is provided in Technical Manual S9310-AQ-SAF-010.

6.19.1 New cells and batteries. All transportation of new lithium cells and batteries in the public domain is controlled by federal law regulating shipment of hazardous materials. The general regulations are stated in 49 CFR 172.101 and 173.185. Any deviation from the methods described in the CFR must be approved before shipment in the form of an "Exemption" by the Office of Hazardous Material Safety Research and Special Programs Administration, U.S. Department of Transportation, Washington, DC 20590.

6.19.2 Used cells and batteries. All transportation of used lithium cells and batteries in the public domain is controlled by federal law regulating the shipment of hazardous materials. The general regulation, as stated in 49 CFR 172.101 and 49 CFR 173.185, permits shipment of waste lithium cells and batteries to a disposal site by motor vehicle only. The transportation of hazardous waste is regulated by 40 CFR 263, which provides for the proper identification of the transporter and manifesting of the waste.

6.20 Storage. Official Navy Guidance for storage is provided in Technical Manual S9310-AQ-SAF-010.

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6.21 Subject term (key word) listing.

Battery management system
Capacity
Charging
Discharging
Electrolyte
Electrochemistry
Gel-polymer lithium-ion
Heaters
Lithium-ion
Lithium polymer
Materials, hazardous
Qualified Products List
Rechargeable lithium
Safety Data Package
Storage, electrical
Systems, electrochemical
Tests, abusive
Tests, safety

6.22 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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TABLE I. Color.

Battery type	Color
Secondary (rechargeable)	SAE AMS-STD-595, color number: 17875

TABLE II. Identification of defects.

Method of inspection	Defect number	Description
Visual	1	Electrical contact surfaces obstructed by insulation, potting, or sealing compounds
Visual	2	Pitting or blow holes in the cell or battery case
Visual	3	Electrolyte leakage
Visual	4	Location and polarity of terminals not as specified
Visual	5	Corrosion
Visual	6	Particles of foreign material in insulation, potting, or sealing compounds
Visual	7	Insulation missing or damaged
Visual	8	Welds contain blow holes, cracks, or slag inclusions
Visual	9	Crazing of glass in glass-to-metal seals
Visual	10	Burrs on cell or battery case
Visual	11	Cell or battery marking not as specified

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TABLE III. Capacity performance requirements.

Requirement Number	Discharge Cutoff Volts & Temperature (see 4.4.1)	Rate of Discharge Factor	Minimum Time to Cutoff Voltage
(1)	20.0 volts at room temperature	1.0C	1 hour
(2)	20.0 volts at 49 °C	1.0C	1.1 hours
(3)	20.0 volts at –18 °C	1.0C	45 minutes
(4)	20.0 volts at room temperature	4.0C	5 minutes

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TABLE IV. Monitoring performance characteristic of the battery.

Data	Range	Display Resolution	Minimum Accuracy	Notes
Cell Voltage (V)	0 to 5	0.01	± 0.005 V	Individual cell
String Voltage (V)	0 to 5	0.1	± 0.005 V	Parallel cells
Pack Voltage (V)	0 to 35	0.05	± 0.1	Voltage across internal cell assembly
Terminal voltage (V)	0 to 35	0.05	± 0.1	Voltage from DC bus
String current (A)	See applicable specification sheet	0.05	± 2% of full scale	+ value = battery charging - value = battery discharging
Pack current (A)	See applicable specification sheet	0.05	± 2% of full scale	Full scale is determined by maximum rating of current sensor.
Cell Balance Delta (V)	0 to 5	0.01	± 0.005 V	Difference between the max and min
String Balance Delta (V)	0 to 5	0.01	± 0.005 V	Difference between the max and min
Capacity (Ah)	0 to 100	0.1	± 0.5	Derivation from other parameters
Temperature	- 55 ºC (-67 ºF) to 105 ºC (222 ºF)	1.0	± 1 ºC	Temperature sensors utilized by the battery pack
SOC	0 to 100 %	1.0	± 10% SOC	100% = fully charged 0% - fully discharged
SOH	0 to 100 %	1.0	± 10% SOH	100% = BOL 80% = EOL

* The minimum accuracy for current measurement may include a DC offset of less than or equal to 200 mA.

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TABLE V. Format for data storage.

Parameter and Header Name	Number of Columns	Format	Units
Date	1	MM/DD/YY	-
Time	1	HH:MM:SS	-
Cell ## Voltage	a	X.XX	Volts
String ## Voltage	b	XX.XX	Volts
Pack Voltage	1	XX.XX	Volts
Terminal Voltage	1	XX.XX	Volts
String ## Current	b	XXX.XX	Amperes
Pack Current	1	XXX.XX	Amperes
Cell Balance Delta for String ##	b	X.XX	Volts
String Balance Delta	1	X.XX	Volts
Capacity	1	XXX.X	Ampere-hours
Temperature ##	c	XXX.X	Fahrenheit
SOC	1	XXX	%
SOH	1	XXX	%
Battery Temp Max	1	XXX.X	Fahrenheit
Battery Temp Min*	1	XXX.X	Fahrenheit
String ## Temp Max*	b	XXX.X	Fahrenheit
String ## Temp Min*	b	XXX.X	Fahrenheit
Battery Discharge Current Max*	1	XXX.XX	Amperes
Battery Charge Current Max*	1	XXX.XX	Amperes
Number of Charge Cycles to Date*	1	XXXX	-
Cumulative Ah Charged*	1	XXX,XXX	Ampere-hours
Cumulative Ah Discharged*	1	XXX,XXX	Ampere-hours

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TABLE VI. Qualification inspection and conformance inspection of batteries.

Test Number	Examination and tests	Qualification inspection							Conformance inspection		
		Sample Number					Requirement Paragraph	Test Paragraph	Group A Sample 1 100%	Group B Same Number	
		1	2	3	4	5				1	2
1	Condition of battery	X	X	X	X	X	3.9.1	4.5.1		X	X
2	Visual and mechanical examination	X	X	X	X	X	3.9.2	4.5.2	X	X	X
3	Dimensions and weight	X	X	X	X	X	3.9.3	4.5.3	X	X	
4	Dielectric strength and insulation resistance	X		X			3.9.4	4.5.5	X	X	
5	Vent tubes	X	X	X	X	X	3.9.5	4.5.6	X	X	X
6	Open circuit voltage	X	X	X	X	X	3.9.6	4.5.4	X	X	X
7	Capacity and electrical performance	X	X	X	X	X	3.9.7	4.5.8	X	X	X
8	High and Low Voltage Charge Test	X					3.9.8	4.5.9		X	
9	Strength of receptacle		X				3.9.9	4.5.10			
10	Cycling	X					3.9.10	4.5.11 Table III		X	
12	Electromagnetic interference				X		3.9.12 3.10.5 3.10.6	4.5.12		-	
13	Discharge while inverted		X				3.6.2	4.5.14			X

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TABLE VI. Qualification inspection and conformance inspection of batteries - Continued.

Test number	Examination and tests	Qualification inspection							Conformance inspection		
		Sample Number					Requirement Paragraph	Test Paragraph	Group A Sample 1 100%	Group B Same Number	
		1	2	3	4	5				1	2
14	Temperature rise and float		X				3.9.13	4.5.15 Table III req 1			X
15	Thermal Protection			X			3.8.2	4.5.16			
16	Physical integrity at high temperature				X		3.8.1, 3.9.11 Table III, req 1	4.5.17			
17	Penetration test				X		3.9.14	4.5.36			
18	Crush test			X			3.9.15	4.5.37			
19	Ground storage					X	3.9.16	4.5.34			
20	Shelf life*					X	3.9.10 3.9.17	4.5.35 4.5.11			
21	Impact resistance				X		3.9.18	4.5.18			
22	Output performance		X				3.9.19	4.5.19			X
23	Discharge and charge at 71ºC				X		3.9.20	4.5.20		X	
24	Cold and hot temperature change				X		3.9.21	4.5.21			X
25	Low Pressure (altitude)		X				3.6.1.2 3.8.1 3.9.3 3.10.5 3.10.6	4.5.23			

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TABLE VI. Qualification inspection and conformance inspection of batteries - Continued.

Test Number	Examination and tests	Qualification inspection							Conformance inspection		
		Sample Number					Requirement Paragraph	Test Paragraph	Group A Sample 1 100%	Group B Same Number	
		1	2	3	4	5				1	2
26	Vibration			X			3.8.1 3.9.3 3.10.5 3.10.6 3.9.11 3.9.22	4.5.24		X	
27	Shock test (basic design)			X			3.8.1 3.10.5 3.10.6 3.9.24	4.5.26		X	
28	Shock test (crash safety)			X			3.9.24	4.5.27			
29	Altitude test		X				3.6.1.2 3.8.1 3.10.5 3.10.6	4.5.28			X
30	Humidity test		X				3.6.1.1 3.8.1 3.9.11 3.10.5 3.10.6	4.5.29			X
31	Salt fog test	X					3.8.1 3.9.11 3.10.5 3.10.6	4.5.30		X	
32	Temperature shock	X					3.8.1 3.9.11 3.10.5 3.10.6	4.5.31		X	

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TABLE VI. Qualification inspection and conformance inspection of batteries - Continued.

Test Number	Examination and tests	Qualification inspection							Conformance inspection		
		Sample Number					Requirement Paragraph	Test Paragraph	Group A Sample 1 100%	Group B Same Number	
		1	2	3	4	5				1	2
33	Transient drop test	X					3.9.23	4.5.25			
34	Battery management system (BMS) requirements	B					3.10.1	4.5.38.1		X	
35	BMS power	B					3.10.2	4.5.38.2			X
36	BMS electronics	B					3.10.3	4.5.38.3			
37	Deep discharge recovery			X			3.10.4	4.5.38.4			X
38	State of charge, state of health, and fault indicator	B					3.10.5	4.5.7 4.5.38.5		X	
39	Built-in test	B					3.10.6	4.5.38.6			X
40	Voltage monitoring	B					3.10.7	4.5.38.7		X	
41	Temperature monitoring	B					3.10.8	4.5.38.8			X
42	Current monitoring	B					3.10.9	4.5.38.9			
43	Balancing	B					3.10.10	4.5.38.10		X	X
44	Short circuit and in-rush current protection	B*					3.10.11	4.5.38.11		X	

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TABLE VI. Qualification inspection and conformance inspection of batteries - Continued.

Test Number	Examination and tests	Qualification inspection							Conformance inspection		
		Sample Number					Requirement Paragraph	Test Paragraph	Group A Sample 1 100%	Group B Same Number	
		1	2	3	4	5				1	2
45	Data storage	B	X	X	X	X	3.10.12	4.5.38.12			X
46	External communications	X					3.10.13	4.5.38.13		X	
47	BMS resources	X					3.10.14	4.5.38.14		X	
48	Maintenance equipment compatibility	X					3.10.15	4.5.38.15		X	X
49	Final examination	X	X	X	X	X	3.8.1 3.9.4	4.5.39 4.5.5		X	X

B: Breadboard battery

*One additional battery may be submitted for the Shelf Life Test.

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TABLE VII. Group B sample size (see 6.2.d).

Inspection lot size	Total number of samples
1 through 20	1
21 through 200	2
201 through 500	4
501 through 1000	6
1001 through 3000	8

TABLE VIII. Broadband random background for vibration.

Frequency (Hz)	Amplitude (g^2/Hz)
10 to 40	0.002 to 0.04 (slope of ±6 dB/octave
40 to 500	0.04
500 to 2000	0.04 to 0.002 (slope of ±6 dB/octave)

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TABLE IX. Fault specifications and BMS response.

Fault Code	Fault (Display) Name	Fault Conditions	BMS Response
01	LOW VOLTAGE	(1) Battery string voltage or pack voltage, either during operation or in open-circuit condition, is below 20.0 volts for over 30 seconds. (2) Battery cell voltage is below the minimum cell voltage reported by the battery manufacturer and accounting for 3.10.4.	(1) Isolate and inhibit discharge of the battery string with the low voltage condition until the fault become inactive. (2) If (1) is not applicable, inhibit discharge of the battery pack until the fault becomes inactive.
02	HIGH VOLTAGE	(1) Battery string voltage or pack voltage exceeds the upper voltage limit of the string or pack, as reported by the battery manufacturing, when operating at normal, abnormal, and emergency operations as specified in MIL-STD-704. (2) Battery cell voltage exceeds the maximum cell voltage reported by the battery manufacturer.	(1) Isolate and inhibit charge of the battery string with the high voltage condition until the fault becomes inactive. (2) If (1) is not applicable, inhibit charge of the battery pack until the fault becomes inactive.
03	HIGH CHG CURRENT	(1) Battery pack charge current exceeds the current limit when operating at normal, abnormal, and emergency operations as specified in MIL-STD-704.	(1) Inhibit charge of the battery pack until the fault becomes inactive.

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TABLE IX. Fault specifications and BMS response – Continued.

Fault Code	Fault (Display) Name	Fault Conditions	BMS Response
04	HIGH DCHG CURRENT	(1) Battery pack discharge current exceeds the current limit when operating at normal, abnormal, and emergency operations as specified in MIL-STD-704. The current shall be capable of satisfying engine start, emergency, and fault clearing load requirements shown in the applicable specification sheet.	(1) Inhibit discharge of the battery pack until the fault becomes inactive.
05	EXTERNAL SHORT	(1) The BMS detects an electrical short external to the battery pack (e.g., caused by aircraft electrical wiring) that is beyond the current ratings associated with engine starting, emergency flight, and fault clearing requirements in accordance with the applicable specification sheet.	(1) Inhibit discharge and charge of the battery pack until the fault becomes inactive.
06	HIGH TEMP	(1) Battery internal temperature measured at specific locations (see application specification sheet) exceeds the maximum operational temperature reported by the battery manufacturer and corresponding to the required maximum operating atmospheric temperature of 71 ºC.	(1) Isolate and inhibit discharge and charge of the battery string with the high temperature condition until the fault becomes inactive. (2) If (1) is not applicable, inhibit discharge and charge of the battery pack until the fault becomes inactive.

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TABLE IX. Fault specifications and BMS response – Continued.

Fault Code	Fault (Display) Name	Fault Conditions	BMS Response
07	LOW TEMP	(1) Battery internal temperature measured at specific locations (see applicable specification sheet) is below the minimum operational temperature reported by the battery manufacturer and corresponding to the required minimum operating atmospheric temperature of -40 ºC.	(1) Isolate and inhibit discharge and charge of the battery string with the low temperature condition until the fault becomes inactive. (2) If (1) is not applicable, inhibit discharge and charge of the battery pack until the fault becomes inactive.
08	IMBALANCE	(1) The difference between the maximum and minimum cell voltage for cells within a string has exceeded 50 mV +/- 5 mV following the balancing function.	(1) Isolate and inhibit discharge and charge of the battery string with the imbalance condition until the fault becomes inactive. (2) If (1) is not applicable, inhibit discharge and charge of the battery pack until the fault becomes inactive.
09	BALANCE HIGH TEMP	(1) The BMS is not able to correct an imbalance or perform normal balancing functions either due to a high temperature within the battery or a high atmospheric temperature.	(1) Isolate and inhibit discharge and charge of the battery string with the balance high temperature condition until the fault becomes inactive. (2) If (1) is not applicable, inhibit discharge and charge of the battery pack until the fault becomes inactive.

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TABLE IX. Fault specifications and BMS response – Continued.

Fault Code	Fault (Display) Name	Fault Conditions	BMS Response
10	HEATER FAIL	(1) The heater blankets(s) or other heating elements in the battery are malfunctioning.	(1) The BMS shall not inhibit any functions of the battery.
11	CV SENSE FAIL	(1) The BMS is either (i) unable to receive a signal from any cell voltage sensors, or (ii) is receiving an erroneous signal from any cell voltage sensor utilized by the battery pack	(1) See applicable specification sheet.
12	PV SENSE FAIL	(1) The BMS is either (i) unable to receive a signal from the pack voltage sensor, or (ii) is receiving an erroneous signal from the pack voltage sensor.	(1) See applicable specification sheet.
13	TV SENSE FAIL	(1) The BMS is either (i) unable to receive a signal from the terminal voltage sensor, or (ii) is receiving an erroneous signal from the terminal voltage sensor.	(1) See applicable specification sheet.
14	TEMP SENSE FAIL	(1) The BMS is either (i) unable to receive a signal from the temperature sensor(s), or (ii) is receiving an erroneous signal from the temperature sensor(s).	(1) See applicable specification sheet.

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TABLE IX. Fault specifications and BMS response – Continued.

Fault Code	Fault (Display) Name	Fault Conditions	BMS Response
15	CHR LIMIT FAIL	(1) The BMS is unable to disable charge current limiting when necessary. (2) The BMS is unable to enable charge current limiting when necessary. (3) The BMS is losses capability to control the charge current limit circuitry.	(1) See applicable specification sheet.
16	DCHG LIMIT FAIL	(1) The BMS is unable to disable discharge current limiting when necessary. (2) The BMS is unable to enable discharge current limiting when necessary. (3) The BMS is losses capability to control the discharge current limit circuitry.	(1) See applicable specification sheet.
17	SYSTEM FAIL	(1) The BMS experiences a firmware failure. (2) The BMS is unable to read/write data to file.	(1) See applicable specification sheet.

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BATTERY, STORAGE, AIRCRAFT
26-VOLT, MAINTENANCE-FREE
ELECTROCHEMISTRY: LITHIUM XXXXXX

Information Required	Example
MIL Part No.	M29595/X-X
Replaces (include replacement data from applicable battery specification sheet)	Replaces D8565/X-X
Maximum Weight	29.0 kg
Military "C" Rated Capacity	24.0 Ah/1.0 HR/24°C/20.0 V
National Stock Number	6140-XX-XXX-XXXX
Contract Number	XXXXX
Lot Code	1209-02
Battery S/N	XXXXX
Manufacturer's Name	XXXXX
Date Battery Placed in Service	*

CAUTION HOT SURFACE **

XXX Fill in applicable information.
* To be filled in by user (see 3.7.3)
** Applies to certain batteries with heaters (see 3.7.5)

FIGURE 1. Battery identification marking.

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MAINTENANCE-FREE BATTERY

Lithium Ion

DO NOT REMOVE COVER

THIS BATTERY SHALL BE SERVICED

ONLY IN AN AUTHORIZED BATTERY SHOP.

THIS BATTERY IS TO BE REMOVED FROM

SERVICE ** YEARS FROM DATE FIRST PLACED

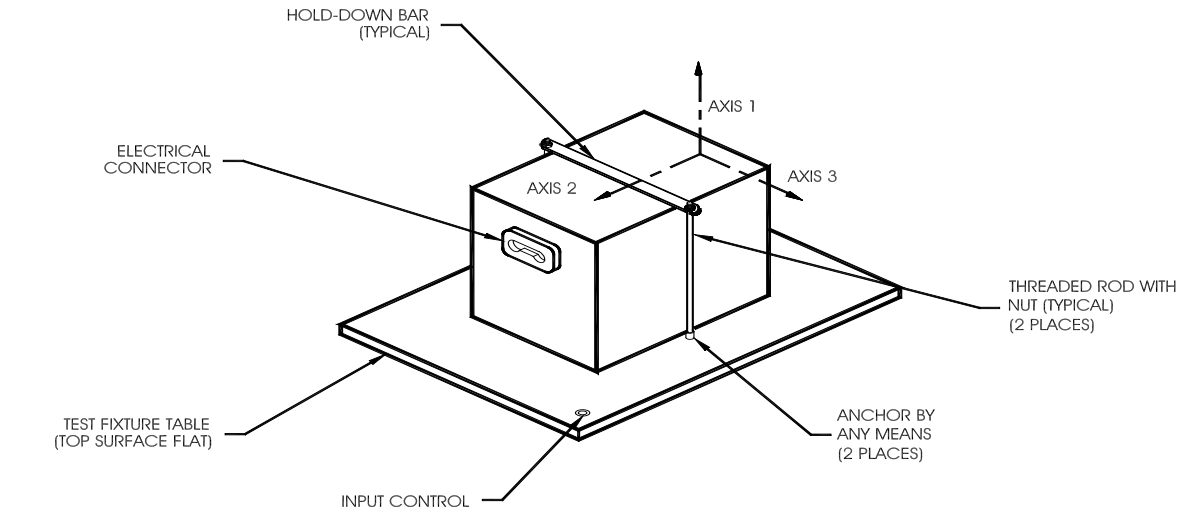
IN SERVICE.

CAUTION: HOT SURFACE ★

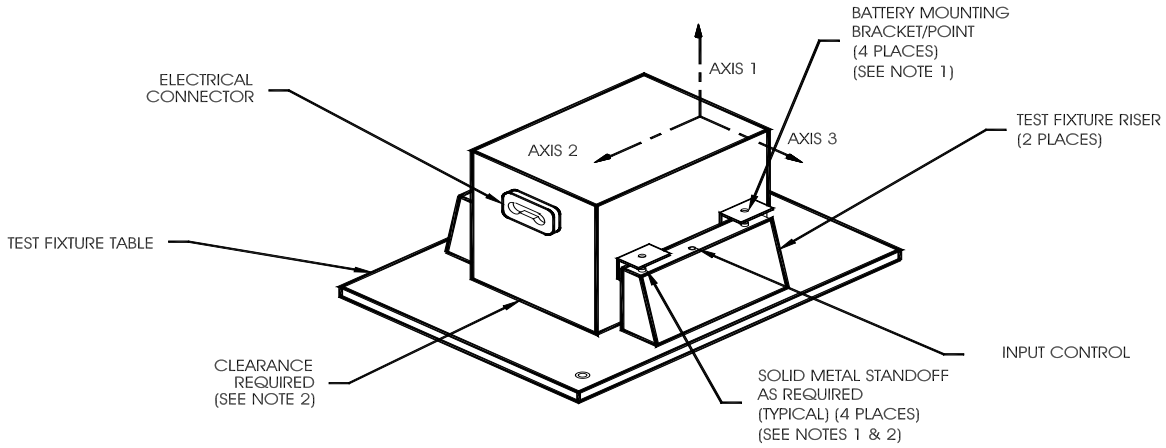
** Fill in information from applicable specification sheet.
★ Applies to certain batteries with heaters (see 3.7.5).

FIGURE 2. Battery caution marking.

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CONVENTIONAL MOUNTED BATTERY

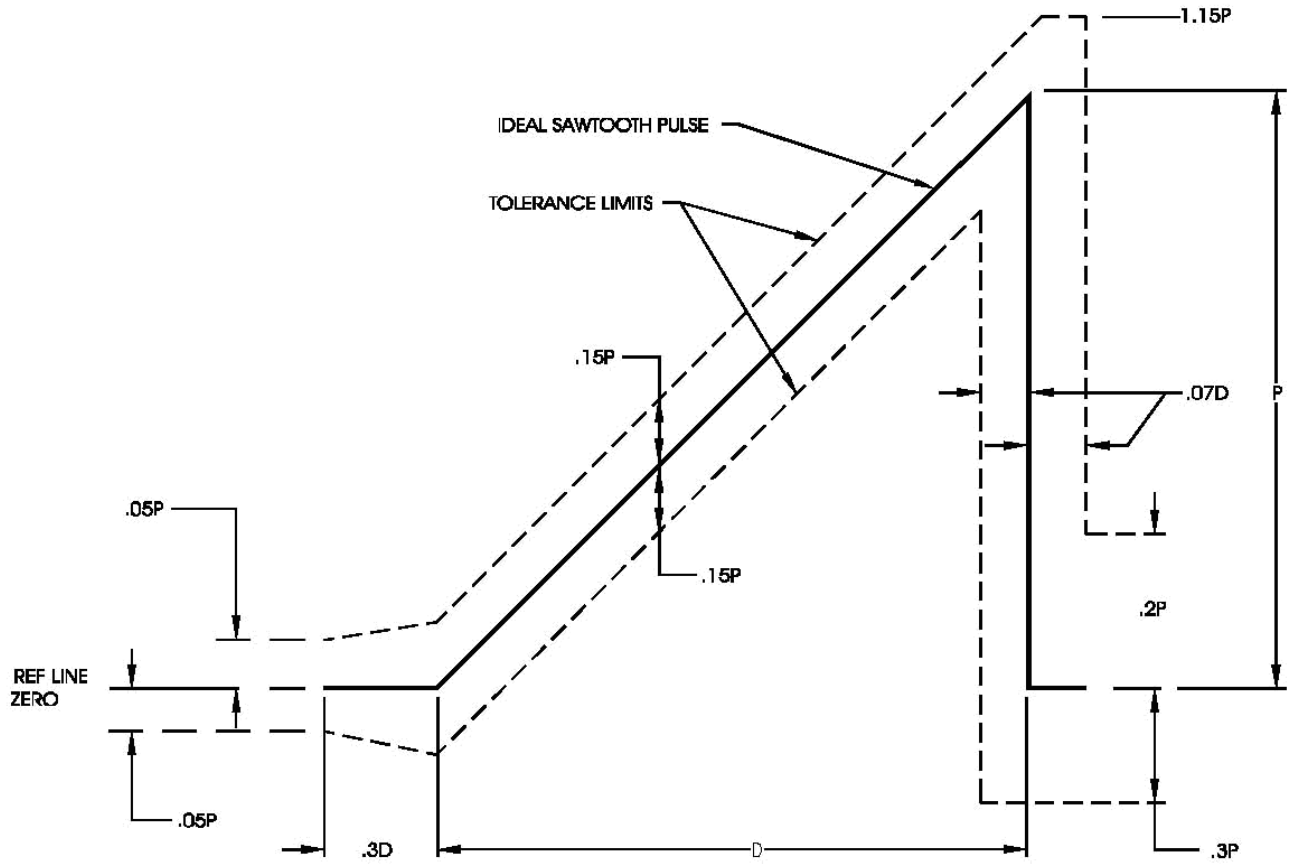


SHOCKMOUNTED BATTERY

NOTES:
1. VIBRATION ISOLATORS SHALL NOT BE USED AT MOUNTING POINTS DURING TESTS SPECIFIED HEREIN.
2. .125 INCH MINIMUM CLEARANCE REQUIRED BETWEEN BOTTOM SURFACE AND TABLE TOP SURFACE.

FIGURE 3. Battery mounting.

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NOTE: The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and its duration is D. The measured acceleration pulse shall be contained between the broken line boundaries and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits of $V_i \pm 0.1\ V_i$, where V_i is the velocity-change associated with the ideal pulse which equals 0.5 DP. The integration to determine velocity change shall extend from 0.4D before the pulse to 0.1D after the pulse.

FIGURE 4. Terminal-peak sawtooth shock pulse configuration and its tolerance limits.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 20
DLA - CC

Review activities:
Army - AV
Air Force - 71

Preparing activity:
Navy - AS

(Project 6140-2020-003)

Agent:
Navy - SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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